Æ 5 B5L E ? p L œ F1b Â(~ –NÎ ø691Š ~

1\ ¼1 "&ó

1 **Ã+X3+5 - =# - = ; G F 8(™**

Ä1 Å2 È3 È3 È4- +b * :" Ä2 Å3- +b * - 4- 2 I * Ê" Ä3 Å3 È3 È- ¼+b * :"

Ä4 Å2 È6- ¼+b *- 3 È6- ¼ ‰ *EË" Ä5 Å2 È5- ¼+b * Ê" Ä6 Å2- +b * - 3- ‰ * !"

Ä7 Å2 È2 È4- 9+b * :" Ä8 Å2- +b * - 3- ‰ * Ê"

2 **ÃB É * ; G F 8(™,´5 ´ ?**

(1) $(CH_3)_3CC(CH_2)_2CH_2CH_3$ (2) $(CH_3)_2CHCH(CH_3)CH_2CH_2CH_2CH_3$

(3) $(CH_3)_3CCH_2CH(CH_3)_2$ (4) $(CH_3)_2CHCH_2C(CH_3)(C_2H_5)CH_2CH_2CH_3$

Ä5 Å$(CH_3)_2CHCH(C_2H_5)CH_2CH_2CH_3$ (6)$CH_3CH_2CH(C_2H_5)_2$

(7) $(CH_3)_2CHCH(CH_3)CH_2CH_3$ (8)$CH_3CH(CH_3)CH_2CH(C_2H_5)C(CH_3)_3$

3 **Ã+•**

4 **Ã ; G 4 F 8(™,´3+5 - =) G Û ² 9KI È 7 *KI X Gü ÛB !".ž - = {** Ä

w 9KI È!".ž - = ² ; Ö

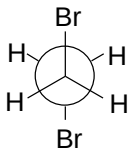
Ä1 Å3- +b * :" Ä2 Å2 È4- ¼+b * !" Ä3 Å3- +b * q 0"

Ä4 Å4- 2 I *EË" Ä5 Å4 È4- ¼+b *EË" Ä6 Å2 È2 È4- 9+b * !"

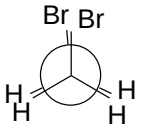
5 **Ã**Ä3 Å Ú Ä2 Å Ú Ä5 Å Ú(1) Ú(4)

6 **Ã+•**

7 **Ã+X4í , Å ¡ ? É * 1 È2- ¼$ä ‰" 00c Ê ú 0 =0c Ê,´ ´B' È ¦ É *B ´B',´ =0 Ä**



交叉式 最稳定



重叠式 最不稳定

8 **Ã** ´B' 2 ´ Ä 1 ÅÈÄ 3 Å ´FP 2 ´ Ä 4 ÅÈÄ 5 Å 1y < Å 2 ÅÈÄ 6 Å

9 Ã 6 €Gÿ j 72 ,´"&ó _ :" ú ¦ 2 ´ *f*

(1) $C(CH_3)_4$ (2) $CH_3CH_2CH_2CH_2CH_3$ (3) $CH_3CH(CH_3)CH_2CH_3$ (4) < (1)

10 **Ã 6 €Gÿ j** 86 **,´"&ó _ !" ú ¦ 2 ´ *f***

(1) $(CH_3)_2CHCH(CH_3)CH_3$ (2) $CH_3CH_2CH_2CH_2CH_2CH_3$, $(CH_3)_3CCH_2CH_3$

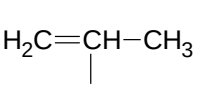
(3)$CH_3CH_2CH(CH_3)CH_2CH_3$ (4)$CH_3CH_2CH_2CH(CH_3)_2$

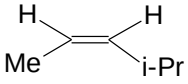
14 **Ã** (4) Ú Ä2 Å Ú Ä3 Å Ú(1)

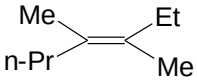
1\ 91 ' &ó

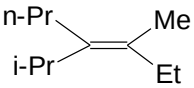
1 Ã+•

2 ÃÄ1 ÅCH_2=CH ² Ä2 ÅCH_3CH=CH ² Ä3 ÅCH_2=$CHCH_2$ ²

(4) 

(5) 

(6) 

(7) 

3Ã 1 Å2- ‰ *-1- :' (2) ý -3,4- ¼+b *-3- Ê' (F(E)-3,4- ¼+b *-3- Ê'

(3) (E)-2,4- ¼+b *-3-"_-3- !' (4) (Z)-1-"O-2-"_-2-$ä-1-.È ‰'

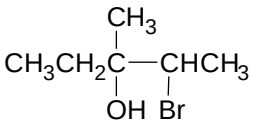
(5) ý -5-+b * -2- Ê' F (E)-5-+b * -2- Ê' (6) ý -3,4- ¼+b *-5- ‰ *-3- Ê'

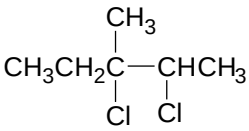
(7) (E)-3-+b * -4- 2 I * -3- Ê' (8) ý -3,4- ¼+b *-3-EË'

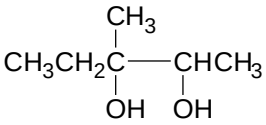
4 Ã+•

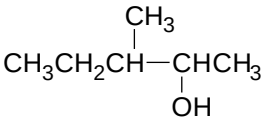
5 Ã+•

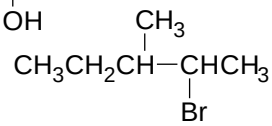
6 Ã

(1) $CH_3CH_2CHCH_2CH_3$ (2) 

(3) 

(4) 

(5) 

(6) $CH_3CH_2COCH_3$ CH_3CHO (7) 

7 Ã#k W]L$ ƒ 6 [j Ö $CH_3CH_2^+$ $CH_3CH^+CH_3$ $(CH_3)_3C^+$

0c Ê W: $CH_3CH_2^+$ Ú $CH_3CH^+CH_3$ Ú $(CH_3)_3C^+$

ý ÄFO Ö: 2 1' Ú I' Ú ‰'

8 Ã+•

9 ÃÄ1 Å$CH_3CH_2CH=CH_2$ (2)$CH_3CH_2C(CH_3)=CHCH_3$ (9Nª Ã ý T/ý)

Ä3 Å$CH_3CH=CHCH_2CH=C(CH_3)_2$ (9 Ã ý T/ý)

+X$KMnO_4$ "W F,´ ×(™: (1) $CH_3CH_2COOH+CO_2+H_2O$ (2)$CH_3CH_2COCH_3+CH_3COOH$

(3) $CH_3COOH+HOOCCH_2COOH+CH_3COCH_3$

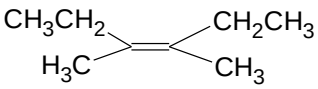
10 ÃÄ1 ÅHBr È E÷"W F(™ Ä2 ÅHBr È 9E÷"W F(™ Ä3 Å 7 H_2SO_4 È 8 H_2O

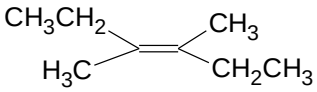
Ä4 ÅB_2H_6/NaOH-H_2O_2 Ä5 Å 7 Cl_2 È500 8 Cl_2 È$AlCl_3$

Ä6 Å 7 NH_3 ÈO_2 8 6Š 8 È E r Ä7 Å 7 Cl_2 È500 È 8 Cl_2 ÈH_2O 9 NaOH

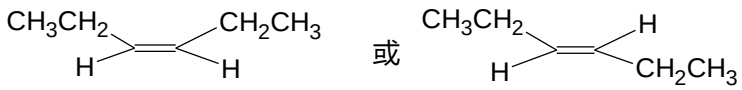
11 Ã' &ó,´5 ´ ? j ÖÄ CH_3 Å$_2$C=$CHCH_3$ Ä 4!• ý Ä ?+•

12 ÃB ' &ó 7-,´5 ´ _ j Ö



或


13 ÃB F 8(™ 7-,´5 ´ ? j Ö



14 Ã$(CH_3)_2C=CHCH_2CH_2C(CH_3)=CHCH_3$ ú $CH_3CH=CHCH_2CH_2C(CH_3)C=C(CH_3)_2$

ú ³ ,´Nª ý 2 ´ *f* .

15 Ã 7 Cl_2 È500 8 Br_2 È

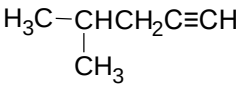
1\ 1 &Ä&ó ¼' &ó

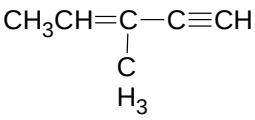
1 Ã+•

2 ÃÄ1 Å2,2- ¼+b *-3- !&Ä (F ‰ * 1 * ‰&Ä) (2) 3-$ä I&Ä (3) 1- 1' -3-&Ä (F ‰' * ‰&Ä)

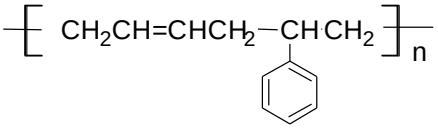
(4) 1- !' -5-&Ä (5) 2-"_-3- !&Ä (6) 4- ‰' * -4- Ê' -2-&Ä (7) (3E)-1,3,5- ! 9'

3 Ã

(1) 

(2) 

(3) $(CH_3)_2CHC\equiv CCH(CH_3)_2$ (4) $HC\equiv CCH_2$ $CH_2C\equiv CH$

(5) $CH_2=CHC(CH_3)=CH_2$ (6) 

Ä7 Å $(CH_3)_3C$ ü & A & & CH_3

4 Ã(1) $CH_3CH_2COOH+CO_2$ (2) $CH_3CH_2CH_2CH_3$ (3) $CH_3CH_2CBr_2CHBr_2$

(4) CH_3CH_2 & A & $ J (5) CH_3CH_2 & A & & X (6)$CH_3CH_2COCH_3$

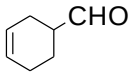
5 ÃÄ1 ÅCH_3COOH Ä2 Å$CH_2CH=CHCH_3$, $CH_3CHBrCHBrCH_3$, CH_3 & A & & +

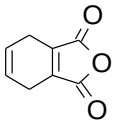
(3) $[CH_3CH=C(OH)CH_3]$, $CH_3CH_2COCH_3$ (4) = ý Ä

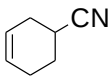
6 Ã(1) $H_2O+H_2SO_4+Hg^{2+}$ (2) 7 H_2, Ç ç ù Ü F r 8 HBr (3) 2HBr

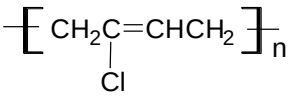
(4) 7 H_2, Ç ç ù Ü F r 8 B_2H_6/NaOH,H_2O_2 (5) 7 f 1-$ä I'' ($H_2$, Ç ç ù Ü F r ;HBr,ROOH) 8 f I&ÄJÐ (Ð $NaNH_2$) 9 $CH_3CH_2CH_2Br+CH_3$ & A & D Q

7 Ã

(1) 

(2) 

(3) 

(4) 

8 Ã(1) $CH_2=CHCH=CH_2$+ & + A & + (2) $CH_2=CHCH=CH_2$ + $CH_2=CHCH=CH_2$

(3) $CH_2CH=C(CH_3)=CH_2$ + $CH_2=CHCN$

9、 (1) ① $HC\equiv CH \xrightarrow[\text{② HBr}]{\text{① } H_2\text{，林德拉催化剂}} CH_3CH_2Br$ （A）

$HC\equiv CH \xrightarrow[\text{液氨}]{NaNH_2} HC\equiv CNa$ （B）

② （A）＋（B） $\longrightarrow CH_3C\equiv CCH_3 \xrightarrow{H_2O，H_2SO_4，HgSO_4} \xrightarrow{\text{还原}}$ TM

(2) +X (1)] f ,´ 1- 1&Ä + 2HCl

(3) +X 1- 1&Ä + H_2, Ç ç ù Ü F r , ½ Ð HBr + ROOH

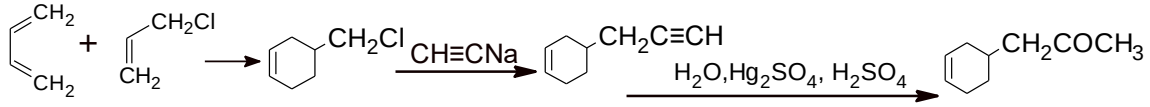
10 Ã

(1) $HC\equiv CH \xrightarrow{NaNH_2} HC\equiv CNa$

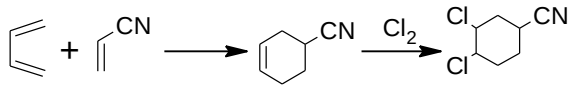
$CH_3CH_2CH=CH_2 \xrightarrow{HBr,ROOR} CH_3CH_2CH_2CH_2Br \xrightarrow{HC\equiv CNa}$

$CH_3CH_2CH_2CH_2C\equiv CH \xrightarrow[HgSO_4]{H_2SO_4,H_2O} CH_3CH_2CH_2CH_2COCH_3$

(2) $CH_3CH=CH_2 \xrightarrow[Cl_2]{500℃} ClCH_2CH=CH_2$ $HC\equiv CH \xrightarrow{Na} CH\equiv CNa$



(3) $CH_3CH=CH_2 \xrightarrow[\text{催化剂}]{NH_3,O_2} CH_2=CHCN$



11 Ã(1) 7 Br_2/CCl_4 8 K&"X$æ#â (2) K&"X$æ#â

12 Ã(1) FJ •.•G¨K&,´"X$æ#â] , ‰&ÄEõFO+O @ ‰&ÄK&"¹#ð6<L" ë

(2) +X Ç ç ù Ü F r ¯ ‰&Ä"R F j ‰' .

13 Ã(1) 1,2- Ð @ , j 1,2- Ð @#k F7- ~ ¾1,4- Ð @#k F7-.

(2) 1,4- Ð @ ×(™" 1,2- Ð @ ×(™,´7-Gÿ ~, $0c Ê .

14 Ã Y+X ´Ø Ë – × f ´¼' Ë – × f *6Aê@ Gú

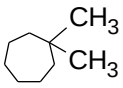
15 Ã!"&ó j : $CH_2=CHCH=CHCH=CH_2$ (9Nª ý 2 ´ *f*)

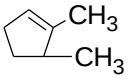
1\ Ä1 72)ß&ó

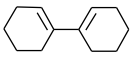
1 ÃÄ1 Å1-+b * -3- 2 I *)ß !' Ä2 Å1-+b * -4- ‰' * -1 È3-)ß ! ¼'

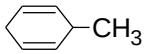
Ä3 Å1 È2- ¼+b *-4- ‰ *)ß :" Ä4 Å3 È7 È7- 9+b * ü)ß [4.1.0] Ê"

Ä5 Å1 È3 È5-)ß Ê 9' Ä6 Å5-+b * ü)ß [2.2.2]-2-EË' Ä7 Å=ê[2.5]-4-EË'

2、 (1) 

(2) 

(3) 

(4) 

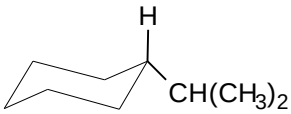
(5)
(6)
(7)

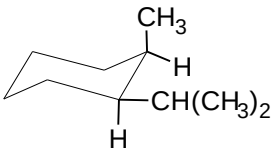
3 Ã+•

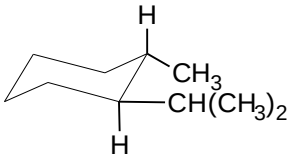
4 ÃÄ1 ÅÃÅ 2 ÅÃÄ 3 Å Ä4 ÅNª Ã ý T/ý Ä5 Å)ß :Nª ý T/ý È üK^ 4Nª ý T/ý È ¡ /ý

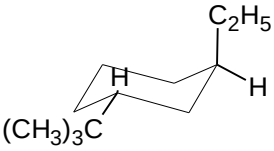
Ä6 Å ˜Nª Ã ý -1 Ã ý -2 Ã ý -4 ¡ /ý

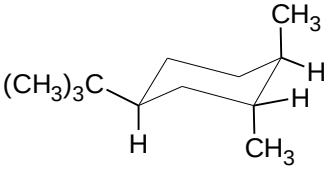
5 Ã

(1) 

(2) 

(3) 

(4) 

(5) 

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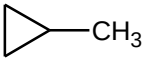
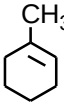
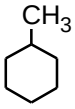
(1) 

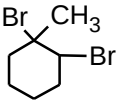
$\xrightarrow{H_2/Ni}$ $CH_3CH_2CH_2CH_3$

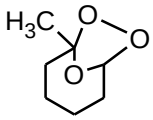
$\xrightarrow{Br_2}$ $CH_3CHBrCH_2CH_2Br$

$\xrightarrow{HI}$ $CH_3CH(I)CH_2CH_3$

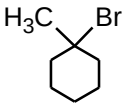
(2) 

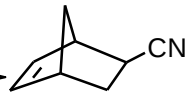
$\xrightarrow{H_2/Ni}$ 

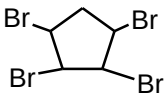
$\xrightarrow{Br_2}$ 

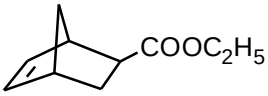
$\xrightarrow{O_3}$ 

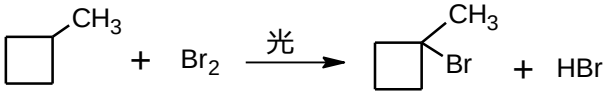
$\xrightarrow{Zn/H_2O}$ $CH_3CO(CH_2)_4CHO$

$\xrightarrow{HBr}$ 

(3)
$\xrightarrow{CH_2=CHCN}$ 

$\xrightarrow{2Br_2}$ 

$\xrightarrow{CH_2=CHCOOC_2H_5}$ 

(4) 

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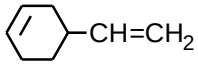

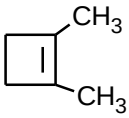

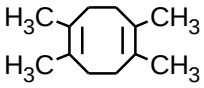

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(A) (B) $CH_3CH_2C{\equiv}CH$ (C) $CH_3C{\equiv}CCH_3$

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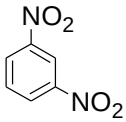
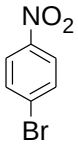
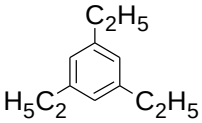
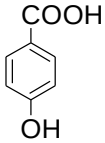
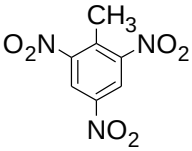
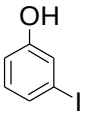
(1) 

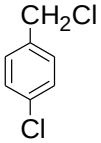
(2) 

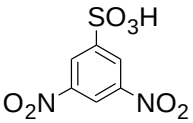
(3) 

(4) 

(5) 

(6) 

(7) 

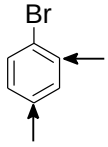
(8) 

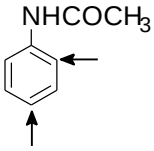
3 ÃÄ1 Å 19 Ä2 Å)"_+b" Ä3 Å).• * ‰9 Ä4 Å8ôG· Ä5 Å9 /*G "_

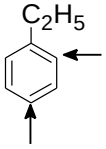
Ä6 Å2 È4- ¼.• *9 +bG¨ Ä7 Å) q ¼" *9 /*G¨JÐ Ä8 Å1-)+b *9 -1- I'

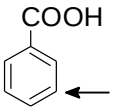
4 ÃÄ1 Å 7 Br_2/CCl_4 8 Ð#ƒ.›G¨ Ä F $HNO_3+H_2SO_4$ Å

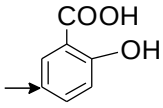
Ä2 Å 7 $Ag(NH_3)_2NO_3$ 8 Br_2/CCl_4

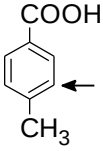
5、(1) 

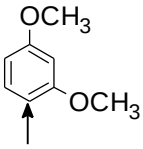
(2) 

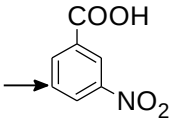
(3) 

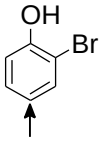
(4) 

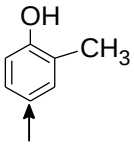
(5) 

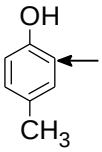
(6) 

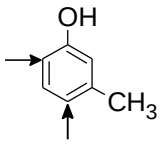
(7) 

(8) 

(9) 

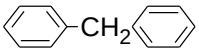
(10) 

(11) 

(12) 

6 ÃÄ1 Å$AlCl_3$ $ClSO_3H$

(2) $CH_3CH=CH_2$

(3) Cl_2,光 

(4)

7 ÃÄ1 ÅA KI È Ä Ç 2 I9 × B KI È Ä j"_ ¢ -H Ä

Ä2 ÅA KI È.• *9 G +O" * F ý Ä × B KI È"W F Ä ÇL$.• *9 +bG¨ Ä

Ä3 ÅA KI È-COCH3 jL$ } Ê } * × B KI È Ä)ß :,´"R Ä

8 ÃÄ1 ÅL$ ¼+b9 Ú) ¼+b9 Ú +b9 Ú 9

Ä2 Å+b9 Ú 9 Ú $ä9 Ú .• *9

Ä3 Å) ¼+b9 Ú +b9 Ú)+b9 +bG¨ Ú)9 ¼+bG¨

Ä4 Å"_9 Ú)"_.• *9 Ú 2 È4- ¼.• *"_9

9 Ã 5 * /j Ä ý Ä ?+• ÅÖ

Ä1 Å4-.• * -2-$ä9 +bG¨ Ö 7 .• F 8 $ä Ä)ß : Å 9 "W F

3-.• * -4-$ä9 +bG¨ Ö 7 $ä Ä)ß : Å 8 "W F 9 .• F

Ä2 Å 7 $ä Ä)ß : Å 8 "W F Ä3 Å 7 "W F 8 .• F Ä4 Å 7 .• F 8 $ä Ä)ß : Å

Ä5 Å 7 .• F 8 "W F (6) 7 " * F 8 "W F 9 "_

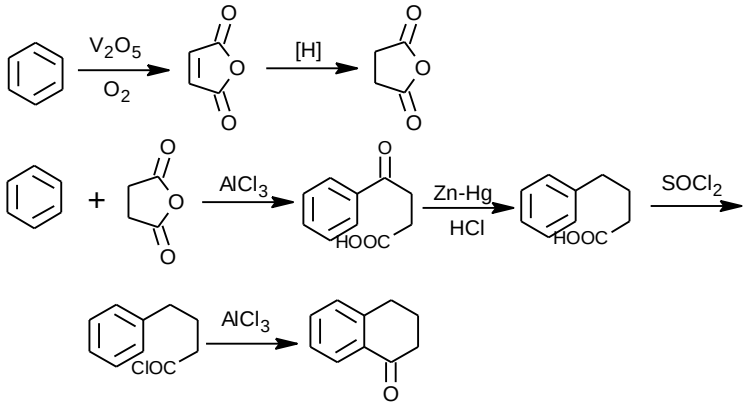
10 Ã 5 * /j Ä ý Ä ?+• ÅÖ

Ä1 Å +b9 j Ï É Ö 7 Cl_2/Fe 8 Cl_2 È y

Ä2 Å 9 j Ï É Ö 7 $CH_2=CH_2$ ÈAlC_3 8 Cl_2 , y 9 9 È$AlCl_3$

Ä3 Å +b9 j Ï É Ö 7 Cl_2 , y 8 CH_3Cl È$AlCl_3$ 9 +b9 È$AlCl_3$

Ä4 Å 9 j Ï É Ö 7 CH_3CH_2COCl ÈAlC_3 8 Zn-Hg+ HCl

(5) 

11 ÃÄ1 Å ‰9 Ä2 ÅL$ ¼+b9 Ä3 Å) ¼+b9 Ä4 Å 2 I9 F!“ I9

Ä5 ÅL$+b ‰9 Ä6 Å w 9+b9

12 Ã T/ý ÖF 9$ä9 × 9/ý Ö • 9$ä9 × 0/ý 0 s.• * F 8(™ Ö w 9$ä9 Ä

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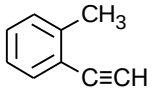
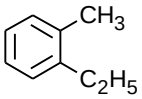
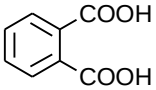
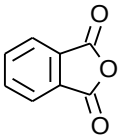
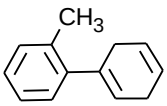
A 

B 

C 

D 

E 

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1 ÃÄ1 Å3 Ä2 Å4 Ä3 Å2 Ä4 Å8 Ä5 Å0 Ä6 Å4 Ä7 Å0 Ä8 Å2 Ä9 Å3

2 Ã+• 3 ÃÄ1Å 4Å 5 Å j { W 6 €

4 Ã) P ƒ Ä 1Å 5 Å -(< Ä 2Å 6 Å MŽ) P ƒ Ä 3Å 4 Å

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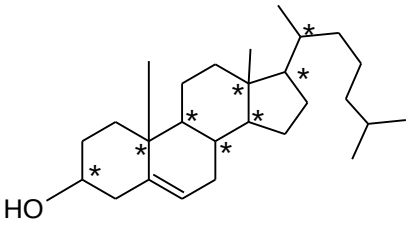
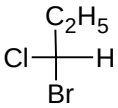
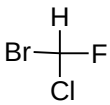
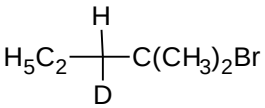
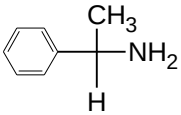
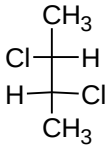
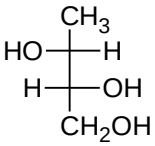
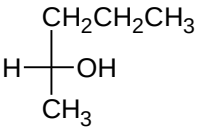
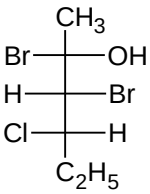
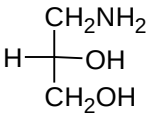
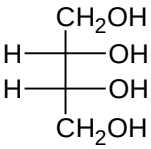
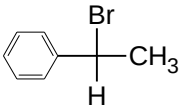
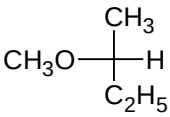


6、(1)  (S) (2)  (S) (3)  (S)

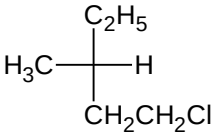
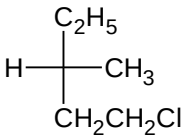
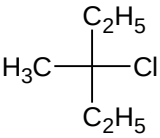
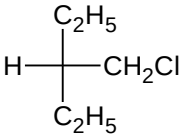
(4)  (R) (5)  (2R,3R) (6)  (2S,3R)

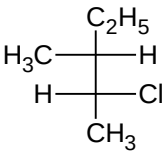
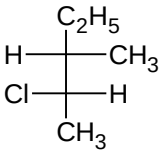
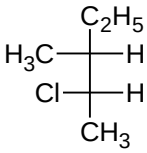
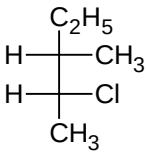
7 ÃÄ4 Å j µ#¸ û ƒ

(1)  (2)  (3) 

(4)  (5)  (6) 

8 Ã

(1)  (2)  (3)  (4) 

(5)  (6)  (7)  (8) 

Ä1Å 2 ÅxÄ 5Å 6 ÅxÄ 7Å 8 Å j) P ƒ

Ä5Å 7 ÅxÄ 5Å 8 ÅxÄ 6Å 7 ÅxÄ 6Å 8 Å jMŽ) P ƒ

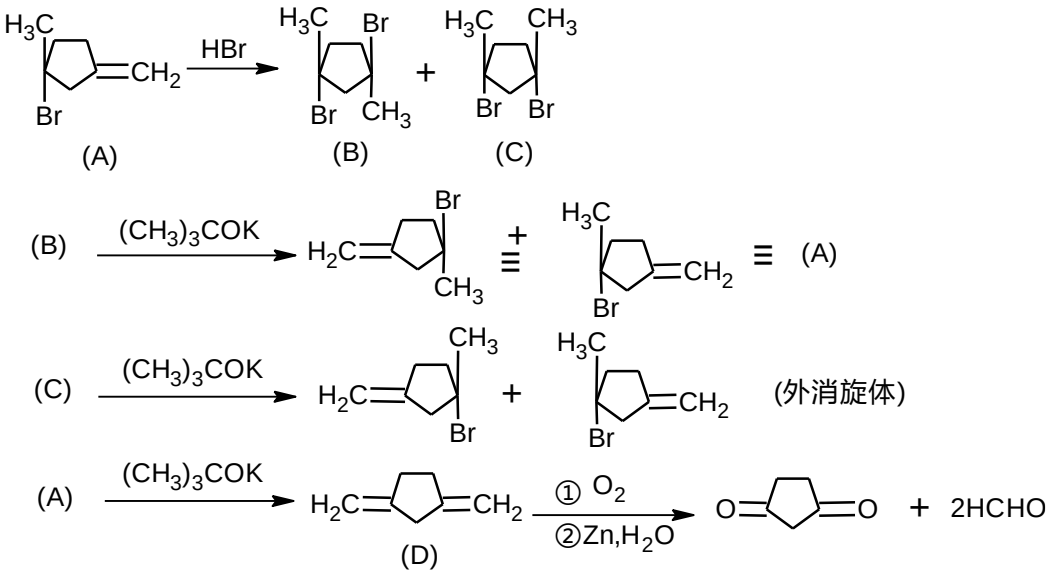
9 ÃÄA ÅCH_3CH_2CH ÄOH Å$CH=CH_2$ ÄB ÅCH_3CH_2CH ÄOH ÅCH_2CH_3

10 ÃÄA Å$CH_2=CHCH_2CH$ ÄCH_3 ÅCH_2CH_3 ÄB Å$CH_3CH=CHCH$ ÄCH_3 ÅCH_2CH_3

10 ÃÄA Å$CH_2=CHCH_2CH$ ÄCH_3 ÅCH_2CH_3 ÄB Å$CH_3CH=CHCH$ ÄCH_3 ÅCH_2CH_3

ÄC ÅCH3CHCH2CH ÄCH3 ÅCH2CH3

11 Ã



1\ •1 ” &ó

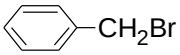
1 ÃÄ1 Å1 È4- ¼"_ 1" Ä2 Å2-+b * -3-"_-6-$ä-1 È4- ! ¼' Ä3Å E Å-2-"_-3- !'

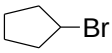
Ä4 Å2-+b * -3- ‰ *-4-$ä :" Ä5 Å)"_$ä9 Ä6 Å3-"_)ß !'

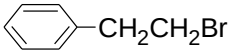
Ä7 Å "O ‰' Ä8 Å4-+b * -1-$ä-1-)ß !'

2 ÃÄ1 Å CH2=CHCH2Cl Ä3 ÅCH3 & A & &ÄÄCH3 ÅCH2Cl

Ä6 ÅCl2C=CH2 Ä7 ÅCF2Cl2 Ä8 ÅCHCl3

(2) 

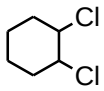
(4) 

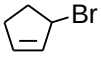
(5) 

3 ÃÄ1 ÅCH3CHBrCH3 CH3CH ÄCN ÅCH3

Ä2 ÅCH3CH2CH2Br CH3CH2CH2OH

Ä3 ÅClCH2CH=CH2 ClCH2CH ÄOH ÅCH2Cl

(4) 

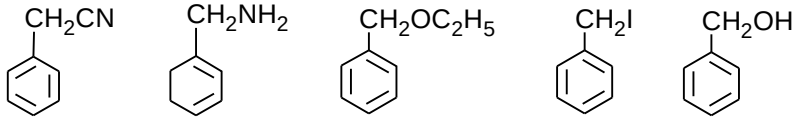
(5) 

Ä6 ÅCH3CH ÄCH3 ÅCHClCH3 CH3CH ÄCH3 ÅCH ÄNH2 ÅCH3

ÄÃÅ CH3 Å2C=CH2 Ä8 ÅPBr3 CH3CH ÄONO2 ÅCH3 $ J % U ;

Ä9 ÅCH3CH2CH2CH2 & A & 0 J ‰C2H5 Ä10 ÅClCH=CHCH2OCOCH3

Ä11 ÅCl2CHCHCl2 Cl2C=CHCl

(12) 

4 ÃÄ 5 * k?± ×(™ È ý Ä ?+• Å

Ä1 Å$CH_3CH_2CH_2CH_2OH$ Ä2 Å$CH_3CH_2CH{=}CH_2$

Ä3 ÅA Ö$CH_3CH_2CH_2CH_2MgBr$ B Ö$CH_3CH_2CH_2CH_3$+ + & A & 0 J % U

Ä4 Å$CH_3CH_2CH_2CH_2I$ + NaBr Ä5 Å$CH_3CH_2CH_2CH_2NH_2$ Ä6 Å$CH_3CH_2CH_2CH_2CN$

Ä7 Å$CH_3CH_2CH_2CH_2ONO_2$ + AgBr Ä8 ÅCH_3 & A & &$_2$G$H_2CH_2CH_3$

Ä9 ÅCH_3 ÄCH_2 Å$_6CH_3$ Ä10 Å$CH_3CH_2CH_2CH_2N$ ÄCH_3 Å$_2$

5 ÃÄ1 Å Ð $AgNO_3$ ÄG· ÅÈ 9,-8¢"¹#ð À * È ý ÄFO Ö Ö

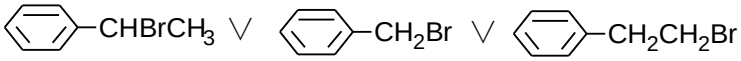
$CH_2{=}CHCH_2Cl$ Ú $CH_3CH_2CH_2Cl$ Ú $CH_3CH{=}CHCl$ Ä ~ = ý Ä Å

Ä2 Å Ð $AgNO_3$ ÄG· ÅÈ 9,-8¢"¹#ð À * È ý ÄFO Ö Ö8ô"_ Ú "_)ß !" Ú "_9 Ä = ý Ä Å

Ä3 Å Ð $AgNO_3$ ÄG· ÅÈ 6 [9,-8¢ Ã<ûTô8¢ ÃTô8¢"¹#ð+O @

Ä4 ÅB r Ã)àB‘ < Ä 2 ÅÈ ý ÄFO Ö Ö8ô"_ Ú 1- 9 * - 2- "_ ‰" Ú "_9 Ä = ý Ä Å

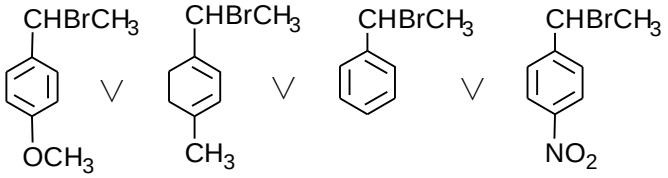
6 ÃÄ1 ÅaÖ CH_3 Å$_3CBr$ Ú $CH_2CH_2CHBrCH_3$ Ú $CH_3CH_2CH_2CH_2Br$

b: 

Ä2 Åa Ö$CH_3CH_2CH_2Br$ Ú ÄCH_3 Å$_2CHCH_2Br$ Ú ÄCH_3 Å$_3CCH_2Br$

b Ö$CH_3CH_2CH_2CH_2Br$ Ú $CH_3CH_2CHBrCH_3$ Ú ÄCH_3 Å$_3CBr$

7 ÃÄ1Å CH_3 Å$_2CBrCH_2CH_3$ Ú ÄCH_3 Å$_2CHCHBrCH_3$ Ú ÄCH_3 Å$_2CHCH_2CH_2Br$

(2) 

8 ÃÄ1 Å$CH_3CH_2CH_2CH_2Br$ ý ÄE³ Ä j+b * _K.,´0ªL$Lk.½L} ~ ¶ S_N2 ý ÄFO Ö Ä

Ä2Å CH_3 Å$_3CBr$ ý ÄE³ È j S_N1 ý Ä ÄÄ CH_3 Å$_2CHBr$ OÆ xF >|S_N2 ý Ä È vB1 j a,´ â h

B r È u ý ÄE³ ’ Ä

Ä3 Å-SH ý Ä ¾ -OH È j S ,´ â h W W ¾ O Ä

Ä4Å CH_3 Å$_2CHCH_2Br$ È j Br " Cl C/ë ë Ä

9 ÃÄ1 ÅS_N2 Ä2 ÅS_N1 Ä3 ÅS_N2 Ä4 Å S_N1 Ä5 Å S_N1 Ä6 ÅS_N2 Ä7 ÅS_N2

10 ÃÄ1Å A ÅKI È$ä Ä Ð X1\ 0 Z.ã ï € : ÄÄ B ÅKI È -OH]#k#, H J > l"?B r ý Ä Ä

Ä2Å A ÅKI È HCl E÷"W F(™ x Ä ÄÄ B ÅKI È ""Fw -CN C#¸L” Ä

Ä3Å B ÅKI È >9)ß-(F ,´$ä =#k#, È = C"d@ Ä

Ä4 ÅKI È Ä#¸L” >9)ß-(Fë.ã :,´"R Ä

11 Ã 5 * /j Ä ý Ä ?+• ÅÖ

Ä1 Å 7 KOH ÄG· Å 8 HBr E÷"W F(™

Ä2 Å 7 KOH ÄG· Å 8 Cl_2 È500 9 [H]

Ä3 Å 7 KOH ÄG· Å 8 Cl_2 9 2KOH ÄG· Å : 2HCl

Ä4 Å 7 KOH ÄG· Å 8 Cl_2 È500 9 Cl_2

Ä5 Å 7 Cl_2 È500 8 HOCl 9 $Ca(OH)_2$,100

Ä6 Å 7 Cl_2 È500 8 Cl_2 9 KOH ÄG· Å : KOH ÈH_2O ; KOH ÄG· Å

Ä7 Å 7 KOH ÄG· Å 8 HCl Ä8 Å 7 KOH ÄG· Å 8 Br_2

Ä9 Å 7 Cl_2 8 H_2/Ni 9 NaCN

Ä10 Å1,1- ¼"_ ‰' Ö 7 HCl 8 Cl_2 9 KOH ÄG· Å

9"_ ‰' Ö 7 2Cl_2 8 KOH ÄG· Å

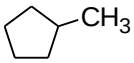
Ä11 Å 7 KOH ÄG· Å 8 Cl_2 È500 9 KOH ÈH_2O

Ä12 Å 7 HCHO/$ZnCl_2$+HCl 8 KOH ÈH_2O

Ä13 Å 7 KOH ÄG· Å 8 Br_2 9 2KOH ÄG· Å : Na È#â"X ; $CH_3CH_2CH_2Br$

12 **Ã+•**

13 **ÃÄ ý Ä ?+• Å**

A B  C

14 **ÃÄ ý Ä ?+• Å**

A: CH_2=CHCH(CH_3)CH_2CH_3 B × CH_3CH_2CH ÄCH_3 ÅCH_2CH_3 Ä û y W Å

15 **ÃÄ ý Ä ?+• Å**

A ÖCH_3CH(Br)CH=CH $_2$ B: CH_3CHBrCHBrCH_2Br C: CH_3CH(OH)CH=CH_2

D: CH_3CH=CHCH_2OH E: CH_3CH(OH)CH_2CH_3 F:$CH_3CH_2CH_2CH_2$OHwsy024

1\ q1 G· ¼GÊ

1 **Ã**Ä1 Å2- :G· 2 e Ä2 Å2- +b * - 2- IG· 3 e Ä3 Å3 È5- ¼+b * - 3- !G· 3 e

Ä4 Å4- +b * - 2- !G· 2 e Ä5 Å1- 1G· 1 e Ä6 Å1 È3- I ¼G· 1 e

Ä7 Å2- IG· 2 e Ä8 Å1- 9 * - 1- ‰G· 2 e Ä9**Å** EÅ- 2- ' - 5- G· 2 e

Ä10 Å2-)ß !' G· 2 e Ä11 Å3- "_ - 1- IG· 1 e

2 **Ã**Ä2 Å Ú Ä 3 Å Ú Ä 1 Å

3 **Ã** 9 ' @"RK^ J A ¼ V ' @"RK^ Â G ÖÄ 4 Å Ú Ä2 Å Ú Ä1 Å Ú Ä3 Å Ú Ä5 Å

4 **Ã**Ä1 Å 7 Br_2 8 $AgNO_3$ (G·)

Ä2 Å+X ' 'B r Ä ZnCl+HCl ÅÈ ý ÄFO Ö Ö3 e Ú 2 e Ú 1 e

Ä3 Å+X' 'B r Ä ZnCl+HCl ÅÈ ¢ -9 ‰G· ý Ä Ä

5 **Ã** 6 [+O @ Ö3-9 * -2- 1G· ¼ 2-+b * :G·

6 **Ã** 5 * k?± ×(™

Ä1 Å$CH_3CH=C(CH_3)_2$ (2) $(CH_3)_2C=CHCH_2OH$ (3) C_6H_5-CH=$CHCH_3$

(4)C_6H_5-CH=$CHCH(CH_3)_2$ (5)$CH_3CH=C(CH_3)C(CH_3)=CHCH_3$

7 ÃÄ1 Å)+b *8ôG· Ú 8ôG· Ú).• *8ôG·

Ä2 Å ¢ -9 * ‰ G· Ú 8ôG· Ú -9 * ‰G·

8 Ã /j Ö X ý ÄE÷0;] È ' @' I *!"/ë € È 6<+O @ T/ý ×(™ Ä

9 Ã+•

10 ÃÄ ý Ä ?+• Å

Ä1 Å$CH_3CH_2CH_2MgBr$ + CH_3CHO F CH_3MgBr + $CH_3CH_2CH_2CHO$

Ä2 ÅCH_3MgBr + $CH_3COCH_2CH_3$ F CH_3CH_2MgBr + CH_3COCH_3

Ä3 ÅCH_3CH_2MgBr + C_6H_5-CHO F C_6H_5-MgBr + CH_3CH_2CHO

Ä4 ÅCH_3MgBr +C_6H_5-$COCH_3$ F C_6H_5-MgBr + CH_3COCH_3

11、(1) $CH_3OH \xrightarrow{[O]} HCHO$

$CH_3CH(OH)CH_2CH_3 \xrightarrow{PBr_3} CH_3CHBrCH_2CH_3 \xrightarrow[\text{干醚}]{Mg} CH_3CH(MgBr)CH_2CH_3$

$\xrightarrow[\text{干醚}]{HCHO} CH_3CH(CH_2OMgBr)CH_2CH_3 \xrightarrow{H_3O^+} CH_3CH(CH_2OH)CH_2CH_3$

(2) $CH_3CH_2CH_2OH \xrightarrow{PBr_3} \xrightarrow[\text{干醚}]{Mg} CH_3CH_2CH_2MgBr$

$CH_2CH(OH)CH_3 \xrightarrow{[O]} CH_3COCH_3 \xrightarrow[\text{干醚}]{CH_3CH_2CH_2MgBr} H_3C-C(CH_3)(OMgBr)-CH_2CH_2CH_3$

$\xrightarrow{H_3O^+} H_3C-C(CH_3)(OH)-CH_2CH_2CH_3$

(3) $HCHO + CH_3CH_2MgBr \xrightarrow{\text{干醚}} \xrightarrow{H_3O^+} CH_3CH_2CH_2OH$

$CH_3CHO + CH_3MgBr \xrightarrow{\text{干醚}} \xrightarrow{H_3O^+} CH_3CH(OH)CH_3$

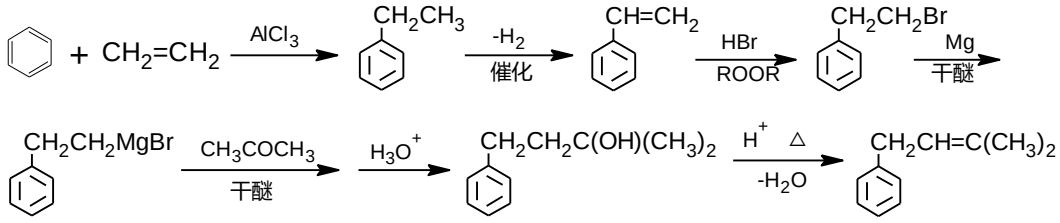
(4) $(CH_3)_2CHOH \xrightarrow{PBr_3} \xrightarrow[\text{干醚}]{Mg} (CH_3)_2CHMgBr$

$(CH_3)_2CHCH_2OH \xrightarrow{[O]} (CH_3)_2CHCHO \xrightarrow[\text{干醚}]{(CH_3)_2CHMgBr} (CH_3)_2CHCH(OMgBr)CH(CH_3)_2$

$\xrightarrow{H_3O^+} (CH_3)_2CHCH(OH)CH(CH_3)_2 \xrightarrow[-H_2O]{H^+ \ \triangle} (CH_3)_2CHCH=C(CH_3)_2$

Ä5 Å 7 Cl_2 È500 8 H_2O È$NaOH$ 9 HOCl : H_2O È$NaOH$; $3HNO_3$

(6) $CH_3CH=CH_2 \xrightarrow{水合} \xrightarrow{氧化} CH_3COCH_3$



12Ã 5 * /j Ä ý Ä ?+• ÅÖ

Ä1 Å 7 ±H_2O 8 HCl Ä2 Å 7 -H_2O 8 -$ Õ FL$ Õ"d 8

Ä3 Å 7 ±H_2O 8 HBr 9 KOH/ G·

13Ã 5 * /j Ä ý Ä ?+• ÅÖ

Ä1 Å 7 PBr_3 8 Mg/ ¢GÊ 9)ß"W ‰" : H_2O

Ä2 Å 7 $CH_3CH_2CH_2Mg$ È ¢GÊ 8 H_3O^+ 9 " H_2O/H È ¸ : .¬"R F"W F

Ä3 Å 7 $C_2H_5Cl/AlCl_3$ 8 NaOH 9 CH_3I : $CH_3CH_2CH_2COCl/AlCl_3$; $LiAlH_4$

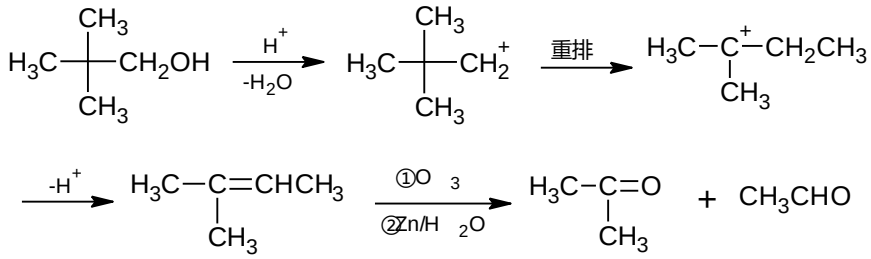
Ä4 ÅF9 1 È3- 1 ¼' ¼ 3- 1' -2-Gž 7 ü' 8 @ 8 CH_3Mg 9 H_3O^+ : H_2/Ni

14 ÃB G·,´5 ´ ? j Ö $(CH_3)_2CHCH(OH)CH_3$

15 Ã Ï F 8(™,´5 ´ ? j : $CH_3CH(OH)CH_2CH_3$ F $CH_3CH_2CH_2CH_2OH$ Ä ý Ä ?+• Å

16 ÃA Ö$(CH_3)_2CHCHBrCH_3$ B Ö $(CH_3)_2CHCH(OH)CH_3$ C Ö$(CH_3)_2C=CHCH_3$

Ä ý Ä ?+• Å

17、 

18ÃA Ö$CH_3CH_2CHBrCH(CH_3)_2$ B Ö$CH_3CH_2CH=C(CH_3)_2$ C Ö$CH_3CH=CHCH(CH_3)_2$

D ÖCH_3CH_2CHO E ÖCH_3COCH_3

Ä 4!• ý Ä ?+• Å

19 ÃÄ1 Å$CH_3OCH_2CH_2CH_3$ +b IGÊ Ä+b"W * I" Å $C_2H_5OC_2H_5$ ‰GÊ Ä ‰"W * ‰" Å

$CH_3OCH(CH_3)_2$ +b 2 IGÊ Ä2-+b"W * I" Å

Ä2 Å$CH_3OCH_2CH_2CH_2CH_3$ +b 1GÊ Ä+b"W * 1" Å

$CH_3OCH(CH_3)CH_2CH_3$ +b " 1GÊ Ä2-+b"W * 1" Å

$CH_3OCH_2CH(CH_3)_2$ +b 2 1GÊ Ä1-+b"W *-2-+b * I" Å

$CH_3OC(CH_3)_3$ +b 1GÊ(2-+b"W *-2-+b * I")

$CH_3CH_2OCH_2CH_2CH_3$ ‰ IGÊ(‰"W * I") $CH_3CH_2OCH(CH_3)_2$ ‰ 2 IGÊ(2- ‰"W * I")

20 ÃÄ1 Å ÐH ŽJÐ È ‰G·JÐ X ‰GÊ] _ * ƒ È 6/ë Ä

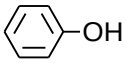
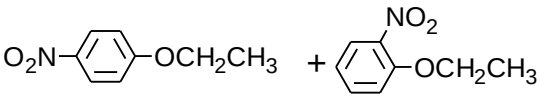
Ä2 Å 7 Ð $Ag(NH_3)_2NO_3$ È1- :&Ä 9,-8¢"¹#ð+O @ È 6/ë È ½ Ð00.•G¨ F Ï j&Ä Ä

8 ÐH ŽJÐ,1 - +b"W *- 3- :G· +O @G·JÐ(* ƒ) È 6/ë È ½ Ð"d F Ï j Ï F 8(™ Ä

21 ÃÄ 5 * k?± ×(™ È ý Ä ?+• Å

Ä1 Å$CH_3OCH_2CH_2CH_3$ + NaI Ä2 Å$CH_3CH_2OCH(CH_3)CH_2CH_3$ + NaBr

Ä3 Å$CH_3CH_2C(CH_3)_2OCH_2CH_2CH_3$ + NaCl Ä4 Å$(CH_3)_2C{=}CH_2$+ $CH_3CH_2CH_2OH$ +NaCl

(5)  + CH_3I (6) 

22 Ã 5 * /j Ä ý Ä ?+• ÅÖ

Ä1 Å f ‰ ¼G· ¼+bGÊ Ö 7 ‰' O_2/Ag È ¸ 8 $2CH_3OH$

f 9+HG· ¼+bGÊ Ö 7 ‰' O_2/Ag È ¸ 8 H_2O ÈH^+

9)ß"W ‰" :)ß"W ‰" ; $2CH_3OH/H_2SO_4$ È ¸

Ä2 Å 7 O_2/Ag È ¸ 8 NH_3 9)ß"W ‰"

Ä3 Å 7 O_2/ CuÈ Ð » Ã ¸ f Ç+bG· 8 ±H_2O

Ä4 Å 7 -$ Õ FL$ Õ"d 8 f 2 IG· 8 ±H_2O

Ä5 Å þ9 * Ö 7 Cl_2/Fe 8 HNO_3+H_2SO_4 9 Na_2CO_3 : CH_3Br

¦] CH_3OH + HBr CH_3Br + H_2O

Ä6 Å 7 +a $CH_2{=}CH_2$ CH_3CH_2MgBr 8 +a $CH_2{=}CH_2$)ß"W ‰"

9 +a 7 ¼ 8 $CH_3CH_2CH_2CH_2OH$: ±H_2O

23 Ã 6 €] [95Ï *Cº J I ' @ 6 €L$"RK^Cº J È"è&éCºQ Ä ‰ ¼G· ¼+bGÊ =7- ' @ 6 €L$,´"R K^ È 6<"è&é _ 965] 0 ~,´ Ä

24 ÃÄ1 ÅCH_3I È$CH_3CH_2CH_2CH_2I$ (2)CH_3I ÈCH_3CH ÄI Å$CH_2CH_2CH_3$

Ä3 ÅCH_3I È$CH_3CH_2CH(CH_3)CH_2I$

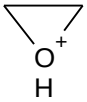
25 ÃB F 8(™,´5 ´ ? j Ö $CH_3CH_2CH_2OCH_2CH_2CH_3$ Ä 9 £ ý Ä ?+• Å

26 Ã!" F 8(™ j Ö $CH_3(CH_2)_4$-O-CH_2CH_3

27 Ãm molAgI = m mol CH_3I F 8(™ $C_{20}H_{21}O_4N$ -( ) 6 €CXGÿ j 339 È

p 11.62mg/235mg(1mol AgI)=11.62/235(1molCH_3I)

$$\frac{m\ mol\ CH_3I}{m\ mol\ 化合物} = \frac{11.62/235}{4.24/339} = 4$$ （此化合物分子中所含OCH_3 数）

28、 (1) CH_3OH +  ⟶ $H_3C{-}OH^{+}{-}CH_2CH_2OH$ $\xrightarrow{-H^+}$ $CH_3OCH_2CH_2OH$

(2) $H_3C{-}O^-$ + ⟶ $CH_3OCH_2CH_2O^-$ $\xrightarrow{+H^+}$ $CH_3OCH_2CH_2OH$

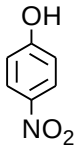
1\ q 01 GŠ ¼G¼

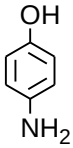
1 ÃÄ1 ÅL$+b9 GŠ Ä2 Å4- ‰ *-1 È3-9 ¼GŠ Ä3 Å2 È3- ¼+b *9 GŠ

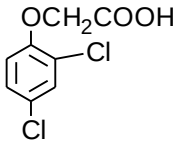
Ä4 Å2 È4 È6- 9.• *9 GŠ Ä5 ÅFë+b"W *9 GŠ Ä6 Å)5Ï *9 /*G¨

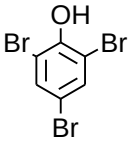
Ä7 Å1-+b * -2-:HGŠ Ä8 Å9-:íGŠ Ä9 Å1 È2 È3-9 9GŠ

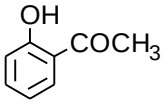
Ä10 Å5-+b * -2- 2 I *9 GŠ Ä11 Å5-.• * -1-:HGŠ Ä12 Å £ -"_:íG¼

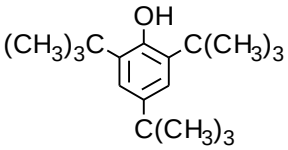
2、 (1) 

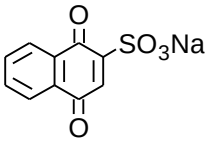
(2) 

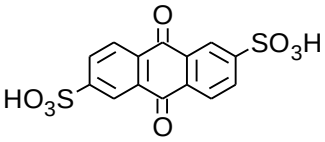
(3) 

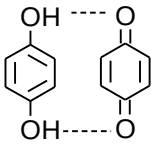
(4) 

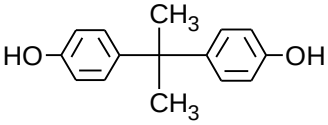
(5) (6) 

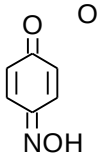
(7) 

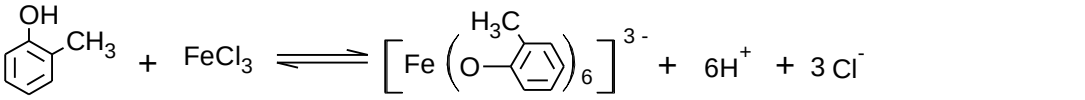
(8) 

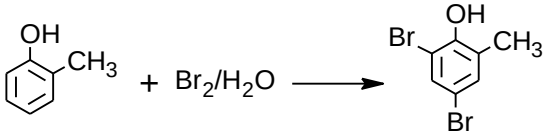
(9) 

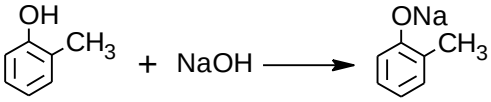
(10) 

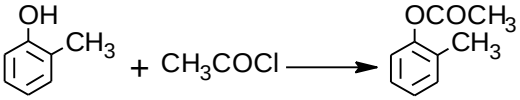
(11) 

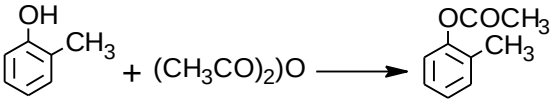
(12) 

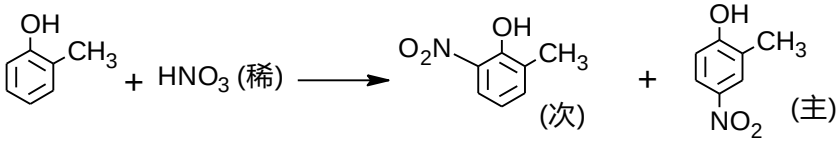
3、 (1) 

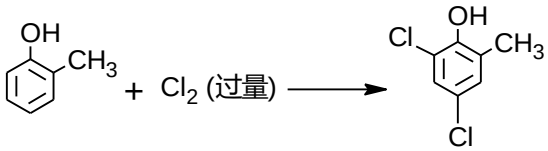
(2) 

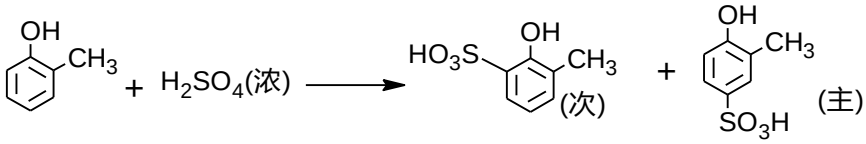
(3) 

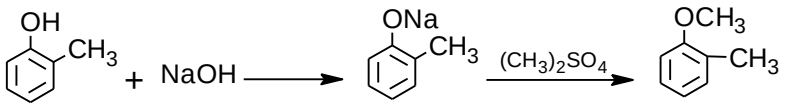
(4) 

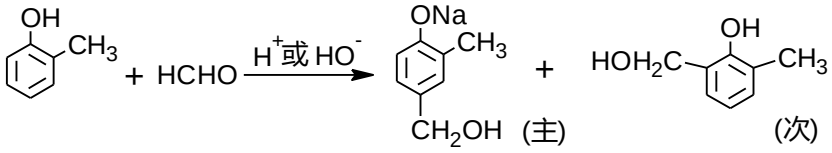
(5) 

(6) 

(7) 

(8) 

(9) 

(10) 

4 Ã 7 $FeCl_3$ 8 Na

5 ÃÄ1 Å+X"R"W FJÐ"d$æ#â È9 GŠ @GŠJÐ$æ ¾"d È'f >+X 6#â%? Ç 6/ë È ½G¨ F Ä

Ä2 ÅÃÄ 3 ÅÃÄ 4 Å < g +X"R"W FJÐ"d$æ#â 6-(Ä ÇGŠ 6/ë * • Ä

6 Ã X9 GŠ 6 €] E • h+e € * ¯G¨ W Î j È¦]Fë Ã) },´G¨ W W ¾L$ } È p G¨ W+a W ` ?,´

Nª ¿ j Ö 2 È4- ¼.• *9 GŠ Ú).• *9 GŠ Ú L$.• *9 GŠ Ú 9 GŠ

7 Ã"d ˜G· =$æ ¾.ãG¨"RJÐ$æ#â6< é ¾"R"W FJÐ$æ#â ÈG¨ F > ø À * È D > 9"_ FJñ$æ#â ý Ä n;

3[8¢ È u Añ > 6 €] [9GŠ5Ï * Ä ƒ+X"R$äG¨ 4*6 È 6 €]5Ï *>Û$ä Ï € È 9 6 r)àB' *)à È

Añ > 6 €] 9G·5Ï * Ä

8 ÃÄ2 ÅÃÄ 4 Å7- ' @ 6 € µ"RK^ È 1 ÅÃÄ 3 Å7- ' @ 6 €L$"RK^ Ä

9 ÃÄ1 Å 9 j Ï É Ö 7 #ƒ.›G¨ Ä/* F Å+O @L$9 9/*G¨ 8 NaOH È ¸ Ä.á'Ä Å 9 H^+

Ä2 Å 9 j Ï É Ö 7 C_2H_5Cl È$AlCl_3$ 8 #ƒ.›G¨ Ä/* F Å+O @ 4- ‰ *-1 È3-9 ¼/*G¨

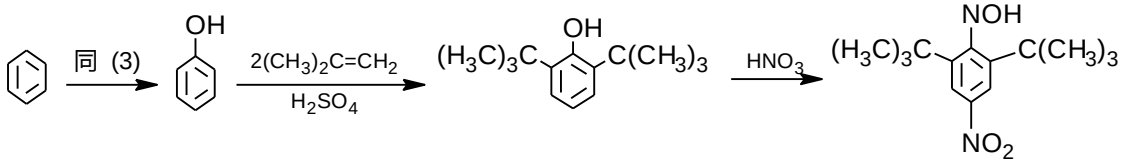
9 NaOH È ¸ Ä.á'Ä Å : H^+

Ä3 Å9 Ö 7/* F È 8 NaOH È ¸ Ä.á'Ä Å 9 H^+ : HNO_2

Ä4 Å+a : f Ç9 GŠJÐ È ½ Ð C_2H_5I £ Ä

Ä5 Å 7 +a : Ç9 GŠ 8 Cl_2 È ¸ 9 Cl_2 È ¸ f Ç 2 È4- ¼"_9 GŠ : NaOH ; $CH_2ClCOOH$

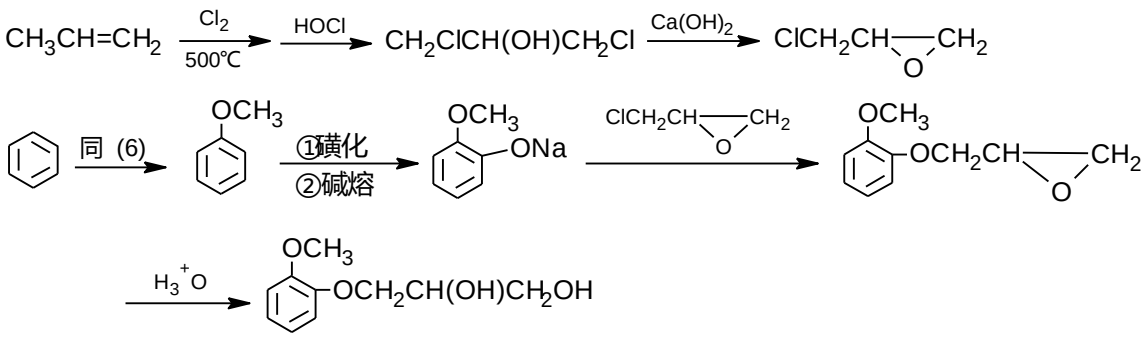
Ä6 Å 7 +a : f Ç9 GŠJÐ 8 CH_3I 9.• F

(8) 

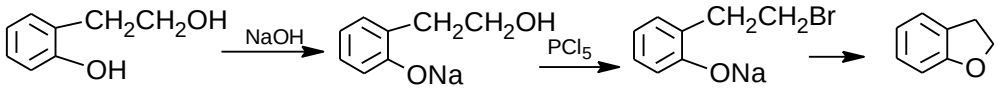
Ä9 Å 7 f 9 GŠ 8 /* F)5Ï *9 /*G¨ 9 Cl_2 : HO ÈH^+ È ¸

Ä10 Å 7 f 9 GŠ 8 C_2H_5Cl È$AlCl_3$ 9 Br_2 È$FeCl_3$

10 Ã

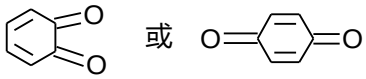


11 ÃÄ1 Å 7 /* F È.á'Ä L$9 ¼GŠJÐ 8 CH_3I

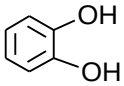
(2) 

Ä3 Å 7/* F È NaOH È ¸ Ä.á'Ä ÅÈ H^+)+b9 GŠ 8 CH_3COCl È $AlCl_3$

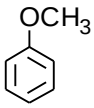
12 ÃB F 8(™5 ´ ? j Ö

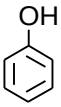
(A) 

或

(B) 

或

13、 (A) 

(B) 

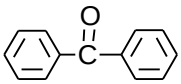
(C) CH_3I

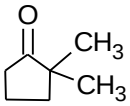
1\ q ¼1 GË ÃGž

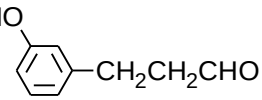
1 ÃÄ1 Å3-+b * :GË Ä2 Å2-+b * -3- :Gž Ä3 Å+b *)ß : *+bGž Ä4 ÅL$+b"W *9 +bGË

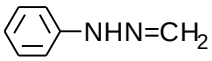
Ä5 Å3 È7- ¼+b *-6-EË' GË Ä6 Å ¢ -$ä 9 ‰Gž Ä7 Å1- :' -3-Gž

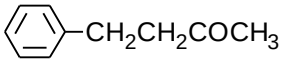
Ä8 Å IGË5Y ¼ ‰G·Ä9 Å)ß !Gž6Ï Ä10 Å2 È4- : ¼Gž Ä11 Å IGž -2 È4- ¼.• *9 7‰

2、 (1) $CH_3CH{=}CHCHO$ (2) 

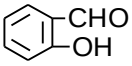
(3) 

(4) 

(5) 

(6) $(CH_3)_2C{=}NNHCONH_2$ (7) 

(8) $CH_3CHBrCHO$ (9)

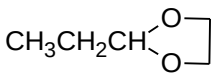
(10) 

3 Ã+•

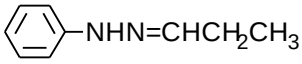
4 ÃÄ1 Å$CH_3CH_2CH_2OH$ Ä2 ÅCH_3CH_2CH ÄOH ÅC_6H_5 Ä3 Å$CH_3CH_2CH_2OH$

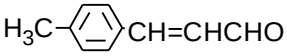
Ä4 ÅCH_3CH_2CH ÄOH ÅSO_3Na Ä5 ÅCH_3CH_2CH ÄOH ÅCN

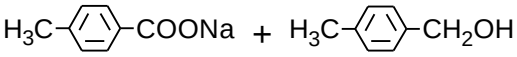
Ä6 ÅCH_3CH_2CH ÄOH ÅCH ÄCH_3 ÅCHO Ä7 Å$CH_3CH_2CH{=}C$ ÄCH_3 ÅCHO

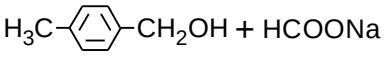
(8) $CH_3CH_2CH_2OH$ (9) 

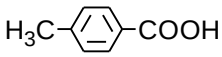
(10) $CH_3CHBrCHO$

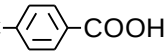
(11) $CH_3CH_2COONH_4$, Ag (12) $CH_3CH_2CH{=}NOH$ (13) 

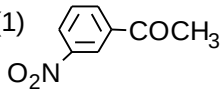
5、 (1) 

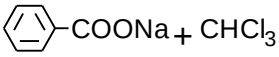
+ H_2O (2) 

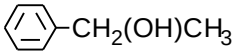
(3) 

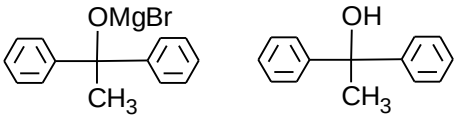
(4) 

(5) 

6、 (1) 

(2) 

(3) 

(4) 

7 ÃÄ1Å 3Å 6Å 7 Å7- +O.È / ý Ä Äx 1Å 2Å 4 Å7- > NaHSO3 +O Ð @ ý Ä È

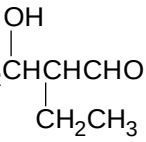
8 ÃÄ1 ÅCF3CHO Ú CH3CHO Ú CH3COCH3 Ú CH3COCH=CH2

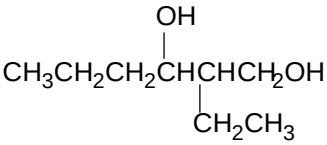
Ä2 ÅClCH2CHO Ú BrCH2CHO Ú CH3CH2CHO Ú CH2=CHCHO

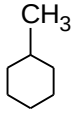
9 ÃÄ1 Å Ð 2 È4- ¼.• *9 6ì Ä2 Å Ð ˆ VB r Ä3 Å.È / ý Ä

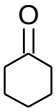
Ä4 ÅO¡ ¼NaHSO3 "d$æ#â Ä5 Å2 È4- ¼.• *9 6ì Ä6 Å.È / ý Ä

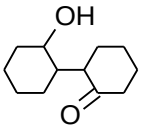
10 Ã 5 * k?± ×(™ È ý Ä ?+• Ö

(1) 

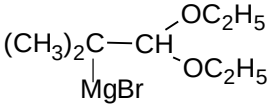
, 

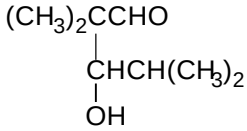
(2) 

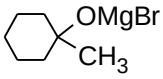
, 

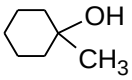
, 

(3) $(CH_3)_2CBrCHO$,

, 

, 

(4) 

, 

, ①H $^+$,△ , ②B $_2H_6$, H_2O_2/OH^-

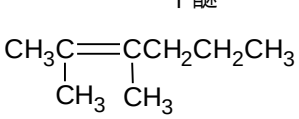
11、 (1) $CH_3CH{=}CH_2 \xrightarrow[\text{过氧化物}]{HBr} CH_3CH_2CH_2Br$

$HC{\equiv}CH \xrightarrow[\text{液氨}]{2NaNH_2} NaC{\equiv}CNa \xrightarrow{2CH_3CH_2CH_2Br} CH_3CH_2CH_2C{\equiv}CCH_2CH_2CH_3$

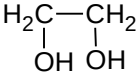
$\xrightarrow[HgSO_4,H_2SO_4]{H_2O} CH_3CH_2CH_2COCH_2CH_2CH_2CH_3$

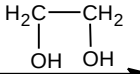
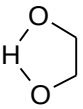
• FJE÷ l"?B r Î.ã Ã"d@ Ã"W F Ç ` Ä

(2) $CH_3CH{=}CH_2 \xrightarrow{HBr} CH_3CHBrCH_3 \xrightarrow[\text{干醚}]{Mg} CH_3CH(MgBr)CH_3$

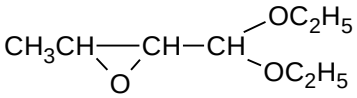
$\xrightarrow[\text{② } H_3O^+,\triangle]{\text{① } CH_3CH_2CH_2COCH_3}$ 

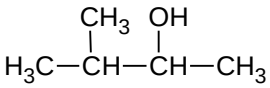
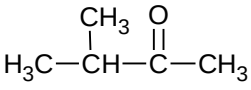
(3) $CH_2=CH_2 \xrightarrow[②\ H_2O]{①\ H_2SO_4} CH_3CH_2OH \xrightarrow{CrO_3/\text{吡啶}} CH_3CHO$

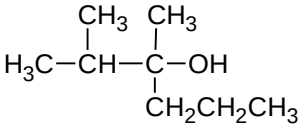
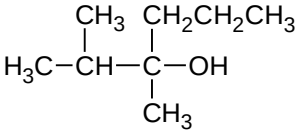
$CH_2=CH_2 \xrightarrow[Ag,\ \triangle]{O_2}$ $\xrightarrow[\triangle]{H_3O^+}$ 

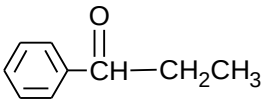
$BrCH_2CH_2CHO \xrightarrow[\text{干}HCl]{}$  $BrCH_2CH_2CH$  $\xrightarrow[\text{干醚}]{Mg}$ $\xrightarrow[②\ H_3O^+]{①\ CH_3CHO} CH_3CH(OH)CH_2CH_2CHO$

(4) $CH_3CH_2OH \xrightarrow{CrO_3/\text{吡啶}} \xrightarrow{\text{稀}\ OH^-} \xrightarrow{\triangle} CH_3CH=CHCHC$

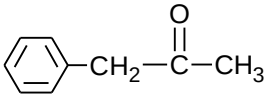
$\xrightarrow[\text{干}HCl]{2C_2H_5OH} \xrightarrow{CH_3COOOH}$ 

12、 A：  B： 

C：  和 

13、 A： 

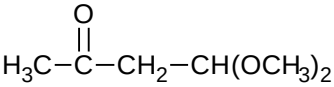
4Ò F yBa1690 cm¹ j5à * h f Ä h.ñ ¡ _ ž ¥1.2 Ä3H Å 9Gý _ ü CH₃ × ¥3.0 Ä2H Å Gý _ ü CH₂ ü × ¥7.7 Ä5H Å JGý j 0 9)ß Ä

B： 

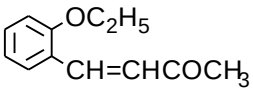
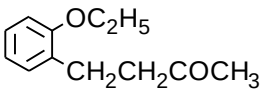
4Ò F yBa1705 cm¹ j5à * h f Ä h.ñ ¡ _ ž ¥2.0 Ä3H Å … _ ü CH₃ × ¥3.5 Ä2H Å … _ ü CH₂ ü × ¥7.1 Ä5H Å JGý j 0 9)ß Ä

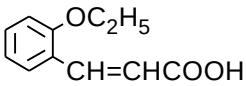
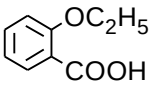
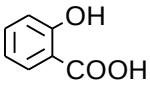
14 ÃB F 8(™5 ´ ? j : $(CH_3)_2COCH_2CH_3$ (ý Ä ?+•)

15 ÃA ÖCH₃COCH₂CH₂CH=C(CH₃)₂ F (CH₃)₂C=C(CH₃)CH₂CH₂CHO B ÖCH₃COCH₂CH₂CHO

16、 

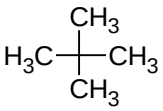
4Ò F yBa1710 cm¹ j5à * h f Ä h.ñ ¡ _ ž ¥2.1 Ä3H Å … _ ü CH₃ × ¥3.2 Ä2H Å JGý _ ü CH₂ ü × ¥4.7 Ä1H Å 9Gý _+b"W *],´ ü CH₃ Ä

17、 A：  B： 

C：  D：  E： 

Ä ý Ä ?+• Å

18、 (1) CH_3OCH_3 (2) $CH_3C{\equiv}CCH_3$ (3) □

(4)  (5) $ClCH_2CH_2Cl$ (6) $(CH_3)_3C-C(CH_3)_3$

19 Ã4Ò F yBa1712 cm¹ j5à * h f È 1383Ã1376 j C ü C ,´ h f Ä h.ñ ¡ _ ž ¥1.00 Ã ¥1.13 _ ü CH_3 × ¥2.13 ü CH_2 ü × ¥3.52 _ CH Ä

A Ö$(CH_3)_2CHCOCH_2CH_3$ B Ö$(CH_3)_2CHCH(OH)CH_2CH_3$

C Ö$(CH_3)_2C{=}CHCH_2CH_3$ D Ö CH_3COCH_3 E ÖCH_3CH_2CHO

(ý Ä ?+•)

1\ q 91 5×G¨ ú ¦>}+O(™

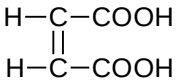
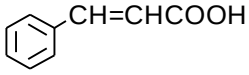
1 Ã(1) !G¨ (2) 2,2,3 - 9+b * 1G¨ (3) 2 -"_ IG¨ (4) £-:H+bG¨ (5) 3 - 1' G¨

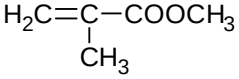
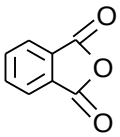
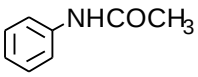
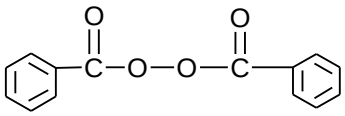
(6))ß !"5×G¨ (7))+b *9 +bG¨+bGŸ (8))9 ¼+bG¨ (9) ¢-:H ‰G¨

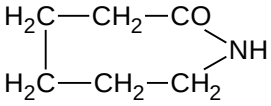
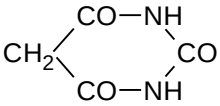
(10) ‰G¨G€ (11) +b * 1' ¼G¨G€ (12) N,N - ¼+b *+bG 7* (13) 3,5 - ¼.• *9 +bG "_

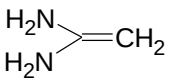
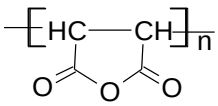
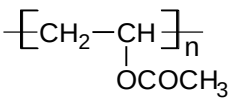
(14) Fë9 ¼+bG Ê7* (15) 2 -+b * -3- 5Ï * 1G¨ (16) 1 -5Ï *)ß :"5×G¨

2 Ã

(1) (2)  (3)  (4) $CH_3(CH_2)_{16}COOH$

(5)  (6)  (7)  (8) 

(9)  (10) $H_2NCOOC_2H_5$ (11) 

(12)  (13)  (14) 

3 Ã+•

4 ÃÄ1 Å9yG¨ Ú I ¼G¨ Ú +bG¨ Ú ‰G¨ Ú 9 GŠ

Ä2 ÅF_3CCOOH Ú $ClCH_2COOH$ Ú CH_3COOH Ú C_6H_5OH Ú C_2H_5OH

Ä3 Å).• *9 +bG¨ Ú L$.• *9 +bG¨ Ú 9 +bG¨ Ú 9 GŠ Ú)ß !G·

5 ÃÄ1 Å 7 $Ag(NH_3)_2OH$ 8 I_2 + NaOH Ä F $NaHCO_3$ Å

Ä2 Å 7 $Ag(NH_3)_2OH$ 8 ¸ Ä3 Å 7 Br_2 8 $KMnO_4$

Ä4 Å 7 $FeCl_3/H_2O$ 8 $NaHCO_3$

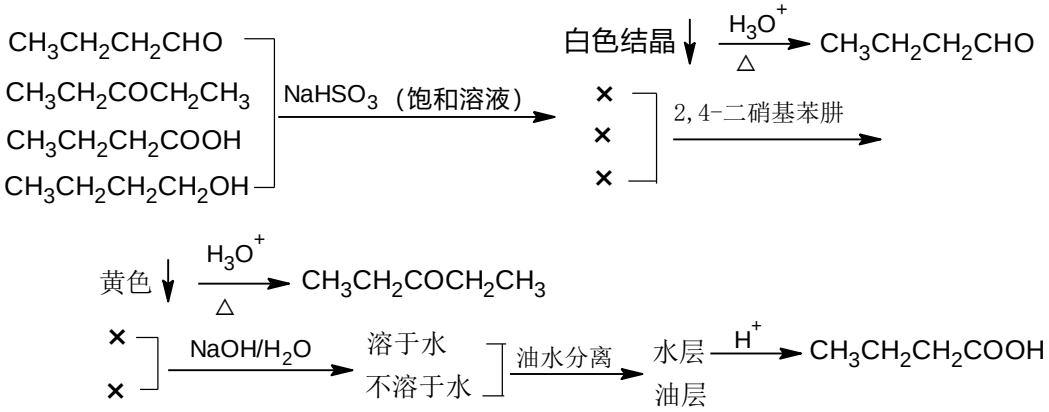
Ä5 Å 7 $AgNO_3$ Ä ‰G "_ 9 AgCl +O @ Å 8 $AgNO_3/C_2H_5OH$

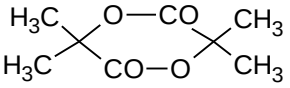
6 Ã(1) $CH_3CBr(CH_3)COOH$ (2) $(CH_3)_2CHCH_2OH$ (3) $(CH_3)_2CHCOCl$

(4) $(CH_3CH(CH_3)CO)_2O$ (5) $(CH_3)_2CHCOBr$

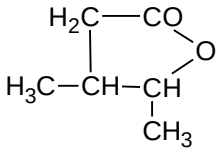
(6) $(CH_3)_2CHCOOC_2H_5$ (7) $(CH_3)_2CHCONH_2$

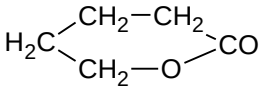
7 Ã



8、(1) 

(2) $CH_3CH{=}CHCOOH$

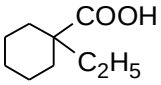
(3) 

(4) 

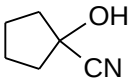
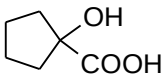
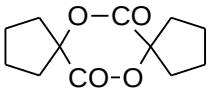
(5) $CO_2 + H_2O$

9 ÃÄ1 ÅA Ö H_3O^+ B ÖPCl_3 ÃPCl_5 Ä F $SOCl_2$ Å C ÖNH_3 D ÖP_2O_5 È ¸

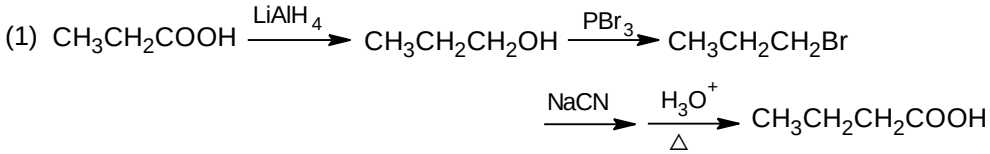
E ÖNH_3 È ¸ F ÖNaOBr + NaOH G ÖH_2/Pd-$BaSO_4$

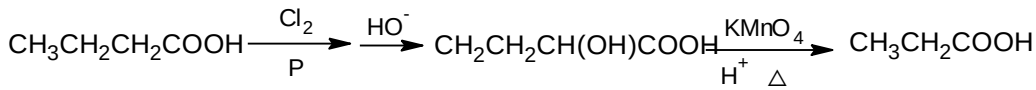
(2) ① C_2H_5MgBr ② H_3O^+, PBr_3 ; 

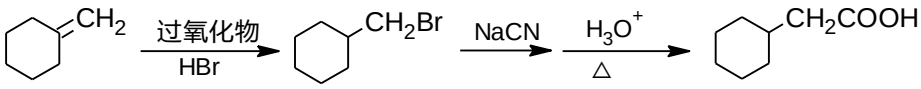
Ä3 Å2NH_3 È$H_2NCONHCONH_2$

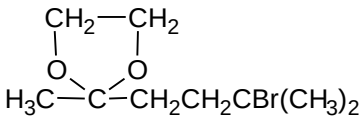
(4)  ;  ; 

10 Ã

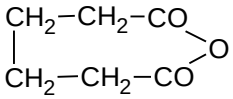
(1) 

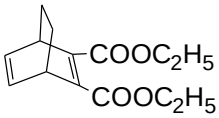
(2) 

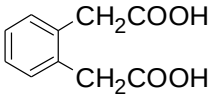
(3) 

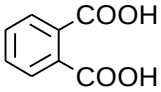
(4) $CH_3COCH_2CH_2CBr(CH_3)_2 \xrightarrow[\text{干HCl}]{HOCH_2CH_2OH}$ 

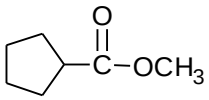
$\xrightarrow[\text{干醚}]{Mg} \xrightarrow{CO_2} \xrightarrow{H_3O^+} CH_3COCH_2CH_2C(CH_3)_2COOH$

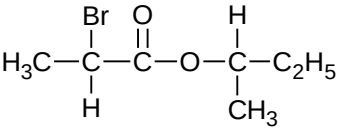
11、 (1) 

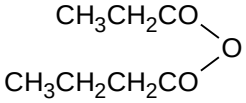
(2) 

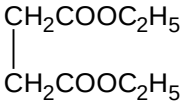
(3) 

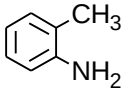
(4) 

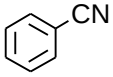
(5) 

(6) 

(7) 

(8) 

(9) 

(10) 

12Ã

(1) $HCOOCH_3$ Ú CH_3COOCH_3 Ú $CH_3COOC_2H_5$ Ú $CH_3COOCH(CH_3)_2$ Ú $CH_3COOC(CH_3)_3$

(2)

13、 (1) $HC{\equiv}CH \xrightarrow[CuCl_2\ \triangle]{HCN} CH_2{=}CHCN \xrightarrow[H_2SO_4]{CH_3OH,\ H_2O} CH_2{=}CHCOOCH_3$

(2) $CH_3CH(OH)CH_3 \xrightarrow{HBr} CH_3CH(Br)CH_3$

$\xrightarrow[\triangle]{NaCN} \xrightarrow{H_3O^+} CH_3CH(CH_3)COOH$

$\xrightarrow[\text{干醚}]{Mg} \xrightarrow{CO_2} \xrightarrow{H_3O^+} CH_3CH(CH_3)COOH$

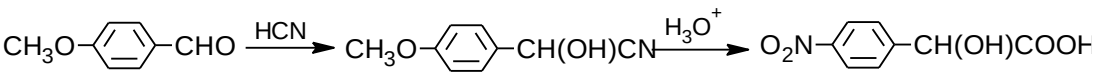
(3) $C_6H_5CH_3 \xrightarrow[\text{光}]{Cl_2} C_6H_5CH_2Cl$

$\xrightarrow[\triangle]{NaCN} \xrightarrow{H_3O^+} C_6H_5CH_2COOH$

$\xrightarrow[\text{干醚}]{Mg} \xrightarrow{CO_2} \xrightarrow{H_3O^+} C_6H_5CH_2COOH$

(4) $CH_3CH_2CH_2COOH \xrightarrow[P]{Cl_2} \xrightarrow{NaCN} \xrightarrow[\triangle]{H_3O^+} CH_3CH_2CH(COOH)_2$

(5) $CH_2{=}CH_2 \xrightarrow{HOBr} HOCH_2CH_2Br \xrightarrow{NaCN} \xrightarrow[\triangle]{H_3O^+} OHCH_2CH_2COOH$

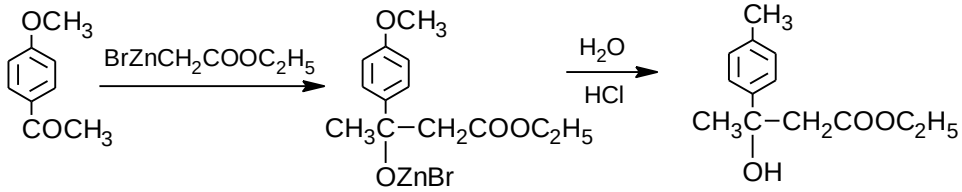
(6) 

(7) $CH_2=CH_2 \xrightarrow{HBr} CH_3CH_2Br \xrightarrow{NaCN} \xrightarrow[\triangle]{H_3O^+} CH_3CH_2COOH \xrightarrow[P]{Br_2} CH_3CH(Br)COOH$

$\xrightarrow[H^+]{C_2H_5OH} CH_3CH(Br)COOC_2H_5 \xrightarrow[干醚]{Zn} CH_3CH(ZnBr)COOC_2H_5 \xrightarrow{CH_3CH_2CHO} \xrightarrow[HCl]{H_2O}$

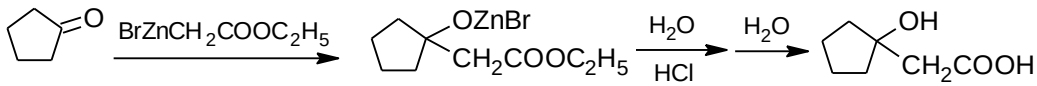
$\xrightarrow{H_2O} CH_3CH(OH)CH(CH_3)COOH$

$CH_3CH_2COOH \xrightarrow{PCl_5} CH_3CH_2COCl \xrightarrow[Pd/BaSO_4]{H_2} CH_3CH_2CHO$

(8) 

$CH_3CH_2OH \xrightarrow{[O]} CH_3COOH \xrightarrow[P]{Br_2} BrCH_2COOH \xrightarrow[H^+ \ \triangle]{C_2H_5OH}$

$BrCH_2COOC_2H_5 \xrightarrow[干醚]{Zn} BrZnCH_2COOC_2H_5$

(9) 

14 Ã þ"_ ` T Z5× *,´D /ë = < ÈB! , x Ä • = < •@ Gú Ä

15 ÃÄ1 Å$CH_3CO_2^{ü}$ CO/ë €,´CO+e9§ £ w 6G} X T Z.ã Ï € : È6< $CH_3CH_2O^{ü}$ CO/ë €,´CO+e9§ Ê X 0 Z"W Ï € : È p E³ =0c Ê È >CX € Œ+X,´ n AE³ W Ä

Ä2 Å$CH_3CH_2CH_2CO_2^{ü}$ Ú $ClCH_2CH_2CO_2^{ü}$ Ä+a ¾Cl ,´B! , Œ+X6<E³0c Ê Å

Ä3 Å$ClCH_2CH_2CO_2^{ü}$ Ú CH_3CH ÄCl Å$CO_2^{ü}$ ÄCl M•F ü$CO_2^{ü}$] ó6<E³0c Ê Å

Ä4 Å$FCH_2CO_2^{ü}$ Ú $F_2CHCO_2^{ü}$ Ä [9 T Z"O Ï € Å

Ä5 Å$CH_3CH_2CH_2CO_2^{ü}$ Ú HC / CCH $_2CO_2^{ü}$ ÄHC / C ü h+e €B! , x Ä Å

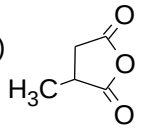
16 Ã10 h 1000h Ä 183.5/1000 Å=1835 { ÈM0 KOH 1.835 s {

17 Ã ý Ä ?+•

ÄA Å,´5 ´ ? j Ö $HOOCCH_2CH(C_2H_5)CH_2COOH$

ÄB Å,´5 ´ ? j Ö $HOOCCH_2CH(C_2H_5)$ COOH

18 ÃÄA Å,´5 ´ ? j Ö $CH_3COOCH=CH_2$ ÄB Å,´5 ´ ? j Ö $CH_2=CHCOOCH_3$

19、 (A) 

(B) CH_2COOH | $CHCOOC_2H_5$ | CH_3

(C) $CH_2COOC_2H_5$ | $CHCOOH$ | CH_3

(D) $CH_2COOC_2H_5$ | $CHCOOC_2H_5$ | CH_3

20 ÃÄ1 Å$HCOOCH_2CH_2CH_3$ Ä2 Å$CH_3CH_2COOCH_3$ Ä3 Å$CH_3COOCH_2CH_3$

1\ q 1 £ ü ¼5à * F 8(™

1 ÃÄ1 Å2 È2 - ¼+b * I ¼G¨ Ä2 Å2- ‰ *- 3- 1GžG¨ ‰GŸ Ä3 Å2- "W)ß !"+bG¨+bGŸ

Ä4 Å+bG "_ * ‰G¨ Ä5 Å3- 1GžG¨ Ä ‰G ‰G¨ Å

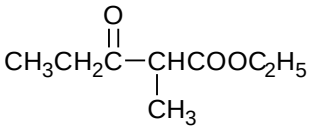
2 ÃÄ1 Å)ß :Gž Ä2 Å$CH_3COCH_2CH_2CH_2COOH$ Ä3 Å$CH_3CH_2CH_2COOH$

3 ÃÄ1 Å Ð $FeCl_3/H_2O$ $CH_3COCH(CH_3)COOC_2H_5$ 9NÌ8¢ ý Ä .

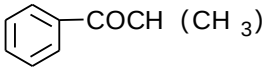
Ä2 Å Ð $FeCl_3/H_2O$ CH_3COCH_2COOH 9NÌ8¢ ý Ä .

4 ÃÄ1 Å Â 2´ Ä2 Å ¡ _ 2´ Ä3 Å Â 2´

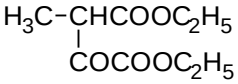
5 Ã

(1) 

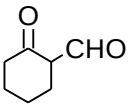
+ C_2H_5OH

(2) 

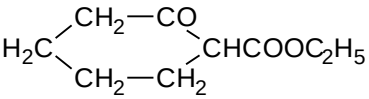
+ C_2H_5OH

(3) 

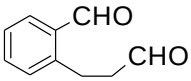
+ C_2H_5OH

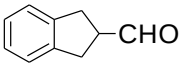
(5) 

+ C_2H_5OH

(4) 

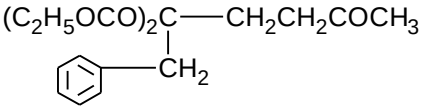
+ C_2H_5OH

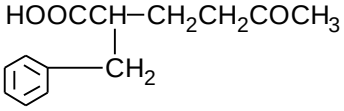
6、(1) 

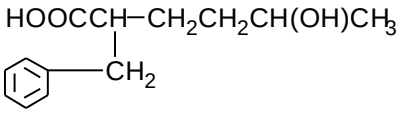
; 

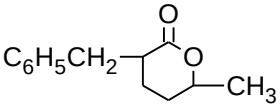
(2) C_2H_5ONa , $CH_3CH(Br)COOC_2H_5$, $CH_3COCH_2CH(CH_3)COOC_2H_5$

(3) $HOCH_2CH_2OH$ / ¢ HCl , $CH_3COCH_2C(OH)(C_6H_5)_2$

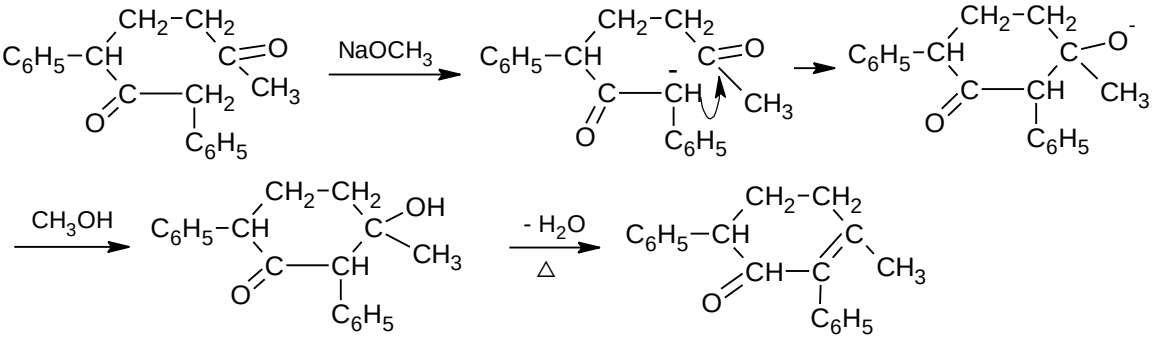
(4) $NaCH(COOC_2H_5)_2$, 

, 



, 

7、 $C_6H_5CH_2COCH_2C_6H_5 \xrightarrow{NaOCH_3} C_6H_5\overset{-}{C}HCOCH_2C_6H_5 \xrightarrow{CH_2=CHCOCH_3}$



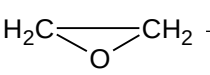
8 Ã I ¼G¨GŸ 8 @ Ö

$$CH_3CH_2OH \xrightarrow{[O]} \xrightarrow[P]{Cl_2} \xrightarrow[②\ H_3O^+]{①NaCN} \xrightarrow[H^+]{2C_3H_5OH} CH_2(COOC_2H_5)_2$$

(1) $CH_3CH_2OH \xrightarrow[H_2SO_4]{NaBr} CH_3CH_2Br$ $\quad CH_3OH \xrightarrow[H_2SO_4]{NaBr} CH_3Br$

$$CH_2(COOC_2H_5)_2 \xrightarrow{C_2H_5ONa} NaHC(COOC_2H_5)_2 \xrightarrow{C_2H_5Br} C_2H_5CH(COOC_2H_5)_2$$

$$\xrightarrow[②\ CH_3Br]{①\ C_2H_5ONa} \xrightarrow[H_2O]{H^+} \xrightarrow[\triangle]{-CO_2} CH_3CH_2CH(CH_3)COOH$$

(2) $CH_3CH_2OH \xrightarrow[\triangle]{H^+} CH_2{=}CH_2 \xrightarrow[Ag]{O_2}$ 

$$CH_3CH_2OH \xrightarrow[H_2SO_4]{NaBr} CH_3CH_2Br \xrightarrow[干醚]{Mg} CH_3CH_2MgBr \xrightarrow{H_3O^+}$$

$$CH_3CH_2CH_2CH_2OH \xrightarrow[H_2SO_4]{NaBr} CH_3CH_2CH_2CH_2Br$$

$$CH_2(COOC_2H_5)_2 \xrightarrow{C_2H_5ONa} \xrightarrow{CH_3CH_2CH_2CH_2Br} \xrightarrow[H_2O]{H^+} \xrightarrow[\triangle]{-CO_2} CH_3CH_2CH_2CH_2CH_2COOH$$

(3) $CH_3CH_2OH \xrightarrow[H_2SO_4]{NaBr} CH_3CH_2Br \xrightarrow{NaCN} CH_3CH_2CN \xrightarrow[H_2O]{H^+} CH_3CH_2COOH$

$$\xrightarrow{LiAlH_4} CH_3CH_2CH_2OH \xrightarrow[\triangle\ -H_2O]{H^+} CH_2{=}CHCH_3 \xrightarrow{Br_2} CH_3CH(Br)CH_2Br$$

$$2CH_2(COOC_2H_5)_2 \xrightarrow{2C_2H_5ONa} 2NaCH(COOC_2H_5)_2 \xrightarrow{CH_3CH(Br)CH_2Br}$$

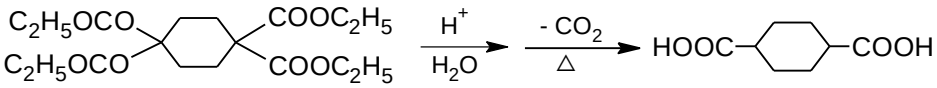
$$\xrightarrow[H_2O]{H^+} \xrightarrow[\triangle]{-CO_2} H_3C-CH(CH_2CH_2COOH)CH_2COOH$$

1,2- ¼$ä ‰'' 8 @Gž (3).

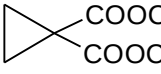
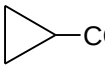
(4) $CH_3CH_2OH \xrightarrow[\triangle\ -H_2O]{H^+} CH_2{=}CH_2 \xrightarrow{Br_2} CH_2BrCH_2Br$

$$2CH_2(COOC_2H_5)_2 \xrightarrow{2C_2H_5ONa} 2NaCH(COOC_2H_5)_2 \xrightarrow{CH_2BrCH_2Br}$$

$$CH_2CH(COOC_2H_5)_2 - CH_2CH(COOC_2H_5)_2 \xrightarrow{2C_2H_5ONa} \xrightarrow{CH_2BrCH_2Br}$$



(5) 1,2- ¼$ä ‰'' 8 @ < (4).

$CH_2(COOC_2H_5)_2 \xrightarrow{2C_2H_5ONa} \xrightarrow{CH_2BrCH_2Br}$  $\xrightarrow[H_2O]{H^+} \xrightarrow[\triangle]{-CO_2}$ 

9 Ã ‰G ‰G¨ ‰GŸ 8 @

$CH_3CH_2OH \xrightarrow{[O]} CH_3COOH \xrightarrow[H^+\ \triangle]{CH_3CH_2OH} CH_3COOC_2H_5 \xrightarrow[H^+]{C_2H_5ONa} CH_3COCH_2COOC_2H_5$

(1) $CH_3CH_2OH \xrightarrow[H_2SO_4]{NaBr} CH_3CH_2Br$

$CH_3COCH_2COOC_2H_5 \xrightarrow[②\ C_2H_5Br]{①\ C_2H_5ONa} \xrightarrow[②\ C_2H_5Br]{①\ C_2H_5ONa} \xrightarrow{5\%NaOH} CH_3CH_2CH(C_2H_5)COCH_3$

(2) $CH_3OH \xrightarrow[H_2SO_4]{NaBr} CH_3Br$

$CH_3COCH_2COOC_2H_5 \xrightarrow[②\ CH_3Br]{①\ C_2H_5ONa} \xrightarrow[②\ CH_3Br]{①\ C_2H_5ONa} \xrightarrow{40\%NaOH} (CH_3)_2CHCOOH$

(3) $CH_3CH_2OH \xrightarrow{[O]} CH_3COOH \xrightarrow[P]{Cl_2} ClCH_2COOH \xrightarrow[H^+]{CH_3CH_2OH} ClCH_2COOC_2H_5$

$CH_3COCH_2COOC_2H_5 \xrightarrow[②\ ClCH_2COOC_2H_5]{①\ C_2H_5ONa} \xrightarrow{5\%NaOH} CH_3COCH_2CH_2COOH$

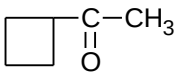
(4) $CH_3CH_2OH \xrightarrow[-H_2O]{H^+} \xrightarrow{Br_2} CH_2BrCH_2Br$

$2CH_3COCH_2COOC_2H_5 \xrightarrow[②\ CH_2BrCH_2Br]{①\ 2C_2H_5ONa} \xrightarrow{5\%NaOH} CH_3COCH_2CH_2CH_2CH_2COCH_3$

(5) $CH_3CH_2OH \xrightarrow[H_2SO_4]{NaBr} CH_3CH_2Br \xrightarrow[干醚]{Mg} CH_3CH_2MgBr$

$CH_3OH \xrightarrow{CrO_3\ /\ 吡啶} HCHO \xrightarrow[②\ H_3O^+]{①\ CH_3CH_2MgBr} CH_3CH_2CH_2OH \xrightarrow[-H_2O]{H^+}$

$CH_3CH{=}CH_2 \xrightarrow[光]{Br_2} \xrightarrow[过氧化物]{HBr} BrCH_2CH_2CH_2Br$

$CH_3COCH_2COOC_2H_5 \xrightarrow[②\ BrCH_2CH_2CH_2Br]{①\ 2C_2H_5ONa} \xrightarrow{5\%NaOH}$ 

10 ÃB F 8(™ j Ö $CH_3COCH_2CH_2COOH$ ý Ä ?+• Ä

11 ÃA: $CH_3CH_2COOC_2H_5$ B: $CH_3CH_2COCH(CH_3)COOC_2H_5$

C: $C_2H_5COC(C_2H_5)(CH_3)COOC_2H_5$ D: $CH_3CH_2COCH(CH_3)CH_2CH_3$

ý Ä ?+• Ä

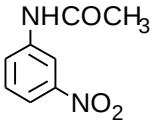
1\ q Ä1 .• * F 8(™ ¼7*

1 Ã Ä1 Å2-+b * -3-.• * :" Ä2 Å I7* Ä3 Å+b 2 I7* Ä4 ÅN- ‰ *L$+b9 7*

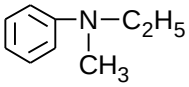
Ä5 ÅN-9 *)9 ¼7* Ä6 Å"R"W F 9+b * 2 IK% Ä7 ÅN-+b *9 /*G 7*

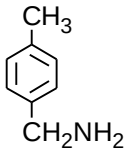
Ä8 Å"_ F 9+b *)$ä9 K% Ä9 ÅN ÈN- ¼+b *) Ê.• * 9 7* Ä10 Å I' 7x

2 Ã

(1) 

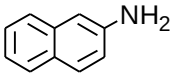
(2) $CH_3NH_2 \cdot H_2SO_4$

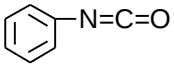
(3) 

(4) 

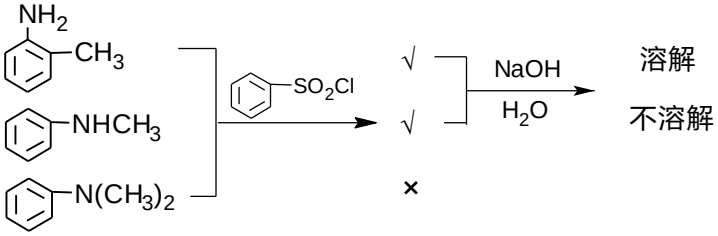
(5) $\begin{array}{l} CH_2CH_2CH_2NH_2 \\ | \\ CH_2CH_2CH_2NH_2 \end{array}$

(6) CH_3NC

(7) 

(8) 

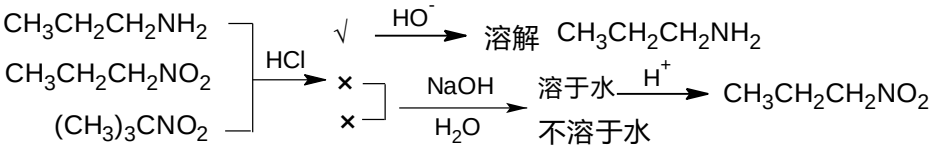
3 Ã(1) 7 $Ag(NH_3)_2OH$ 8 $CHCl_3$/KOH (27x ý Ä) 9 $NaHCO_3$ $æ#â

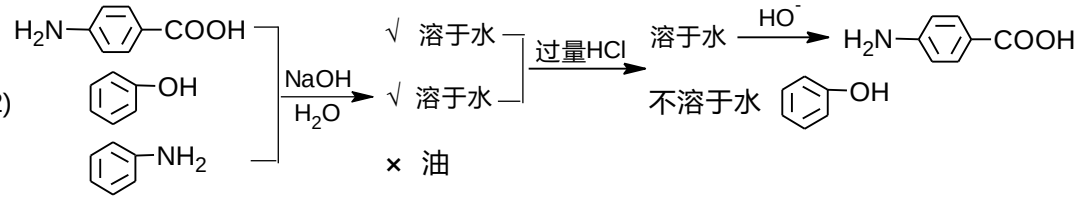
(2) 

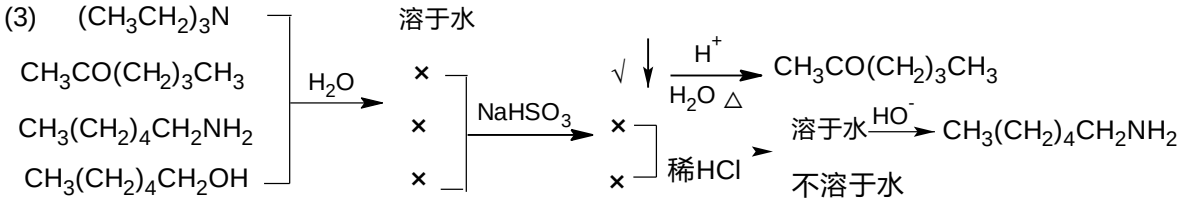
(3) $CHCl_3$/KOH (27x ý Ä)

(4) Br_2/H_2O (F+X%2,-2¹$æ#â È9 7* n3[8¢)

4 Ã

(1) 

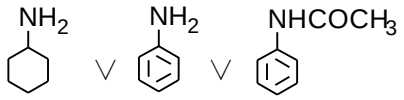
(2) 

(3) 

5 ÃÄ1 Å+b7* Ú "X Ú 9 7* Ú ‰G 7*

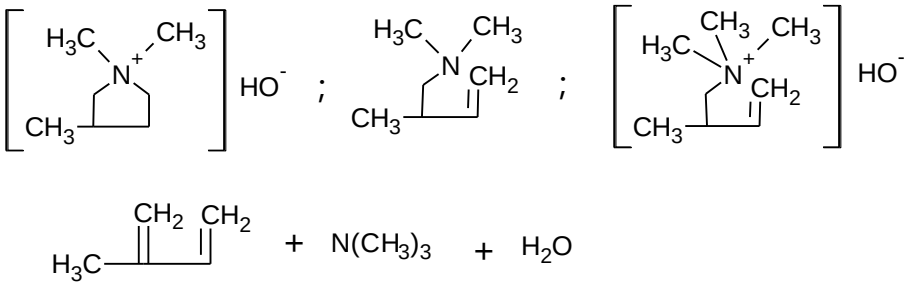
Ä2 Å8ô7* Ú)+b9 7* Ú).• *9 7* Ú 2 È4- ¼.• *9 7*

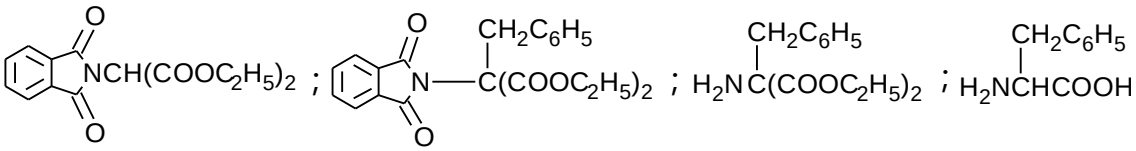
Ä3 Å+b7* Ú N- +b *9 7* Ú 9 7* Ú 99 7*

(4) 

(5)

6 Ã Ä1 ÅCH_3CH_2COOH , CH_3CH_2COCl , $CH_3CH_2CON(CH_2CH_2CH_3)_2$, $(CH_3CH_2CH_2)_3N$

(2) 

(3) 

7 Ã

(1) $(CH_3)_2CHCH_2CH_2OH \xrightarrow{[O]} (CH_3)_2CHCH_2COOH \xrightarrow[②\ NH_3]{①\ SOCl_2} (CH_3)_2CHCH_2CONH_2 \xrightarrow[NaOH]{Br_2} (CH_3)_2CHCH_2NH_2$

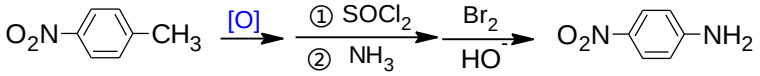
(2) $(CH_3)_2CHCH_2CH_2OH \xrightarrow{PCl_3} (CH_3)_2CHCH_2CH_2Cl \xrightarrow{NH_3} (CH_3)_2CHCH_2CH_2NH_2$

(3) $(CH_3)_2CHCH_2CH_2OH \xrightarrow{PCl_3} (CH_3)_2CHCH_2CH_2Cl \xrightarrow{NaCN} (CH_3)_2CHCH_2CH_2CN \xrightarrow{H_2\ ,\ Ni} (CH_3)_2CHCH_2CH_2CH_2NH_2$

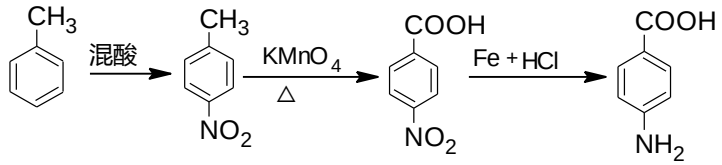
(4) $CH_2{=}CH_2 \xrightarrow{Br_2} BrCH_2CH_2Br \xrightarrow{2NaCN} \xrightarrow{H_2\ ,\ Ni} H_2NCH_2CH_2CH_2CH_2NH_2$

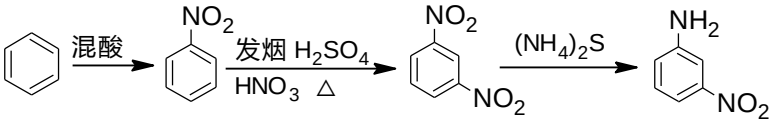
(5) $CH_2{=}CH_2 \xrightarrow{HBr} CH_3CH_2Br \xrightarrow{NaCN} CH_3CH_2CN$

(6) $CH_3CH{=}CH_2 \xrightarrow{Br_2} \xrightarrow{2NaCN} \xrightarrow[\triangle]{H_3O^+} HOOCCH_2CH(CH_3)COOH$

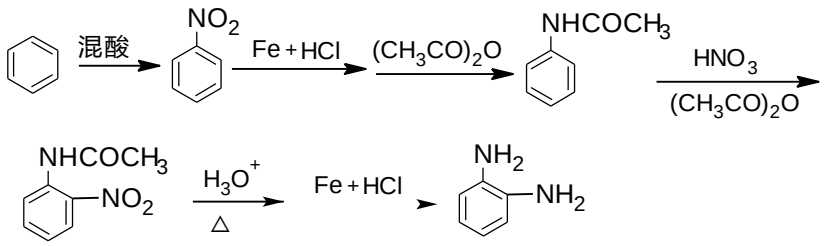
(7) 

8 Ã

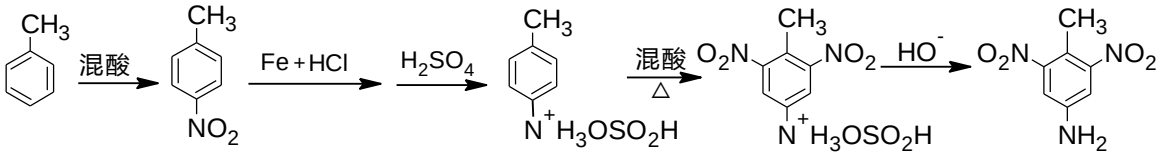
(1) 

(2) 

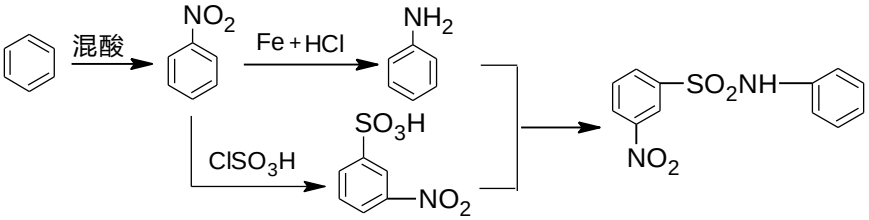
(3)


混酸
NO2
Fe＋HCl
(CH3CO)2O
NHCOCH3
HNO3
(CH3CO)2O
NHCOCH3
NO2
H3O+
△
Fe＋HCl
NH2
NH2

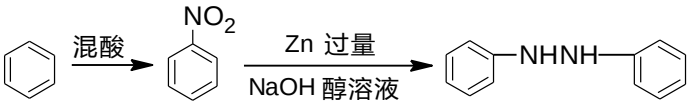
(4)


CH3
混酸
CH3
NO2
Fe＋HCl
H2SO4
CH3
N+H3OSO2H
混酸
△
O2N
CH3
NO2
N+H3OSO2H
HO-
O2N
CH3
NO2
NH2

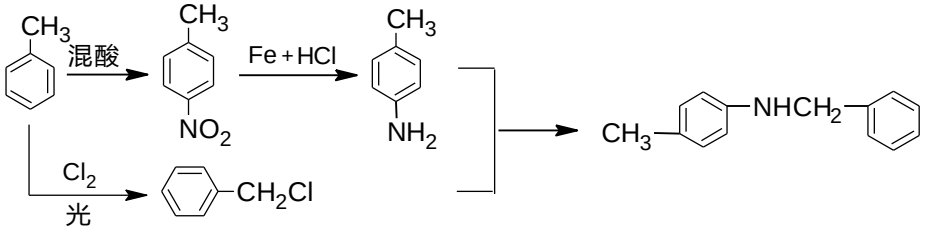
(5)


混酸
NO2
Fe＋HCl
NH2
ClSO3H
SO3H
NO2
SO2NH
NO2

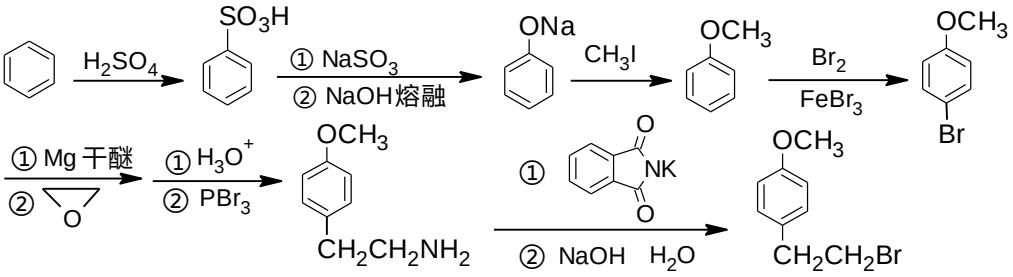
(6)


混酸
NO2
Zn 过量
NaOH 醇溶液
NHNH

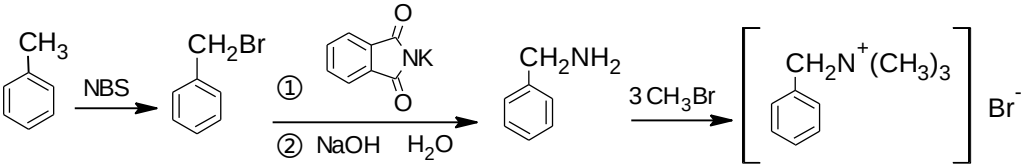
(7)

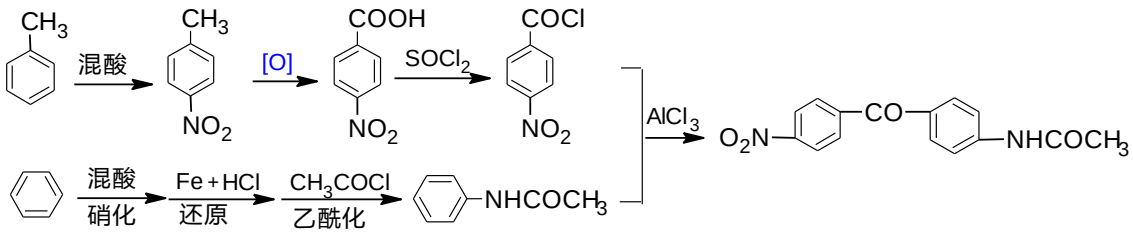

CH3
混酸
CH3
NO2
Fe＋HCl
CH3
NH2
Cl2
光
CH2Cl
CH3
NHCH2

(8)


H2SO4
SO3H
① NaSO3
② NaOH熔融
ONa
CH3I
OCH3
Br2
FeBr3
OCH3
Br
① Mg 干醚
②
O
① H3O+
② PBr3
OCH3
CH2CH2NH2
①
O
NK
O
② NaOH H2O
OCH3
CH2CH2Br

(9)

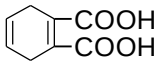

CH3
NBS
CH2Br
①
O
NK
O
② NaOH H2O
CH2NH2
3 CH3Br
CH2N+(CH3)3
Br-

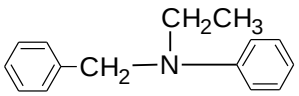
(10) 

9 ÃÄ3 Å,´ 8 @D 4ï 0 8*6 Ä

10 Ã+•

11 ÃA ÖCH2=CHCH2NH2　　B ÖCH3CH2CH2CH2NH2　　C Ö[CH2=CHCH2CH2N(CH3)3]+ I ¨

D ÖCH2=CH ü CH=CH2　　E: 

12 ÃB F 8(™ j Ö 

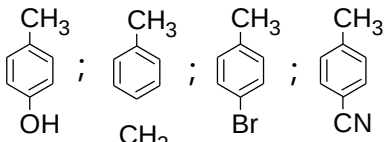
1\ q •1　　Gý"^ F 8(™ ¼ ¦"^ F 8(™

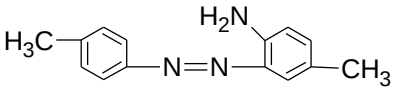
1 Ã(1) Gý"^9 .›G¨-　　(2)) ‰G *Gý"^9 - G¨-　　(3) 4+b * -4 ÿ -5Ï * ¦"^9

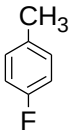
(4) 4-(N,N- ¼+b7* *)-4 ÿ -+b * ¦"^9　　(5) 2,2ÿ - ¼+b *"R F ¦"^9　　(6) ¼"_.ã'

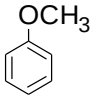
2 Ã Î j Ä j9 *Gý"^- _ 0/ý a,´ â+eB r È u ƒGý"^ *Fë } F) } :F 9 j h+e € *.• * & È
Î j ¶Gý"^- ,´ â+e W È ¯ ¦ 8 ý Ä#k W Q Ä

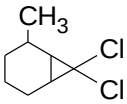
3 ÃGý"^- >9 7* ¦ 8 X aG¨] 9 Y ÈGý"^- >GŠ ¦ 8 X a.á] 9 Y Ä

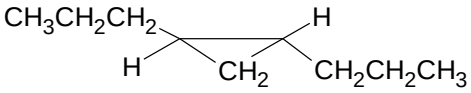
4、 (1) Fe，HCl； $NaNO_2$，HCl； 

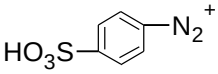
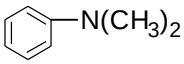


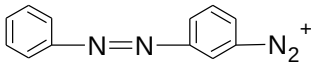
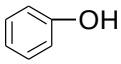
； 

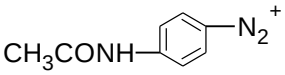
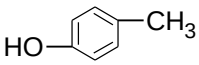
(2) 

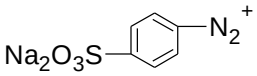
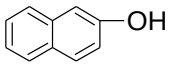
(3) 

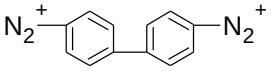
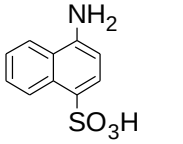
(4) 

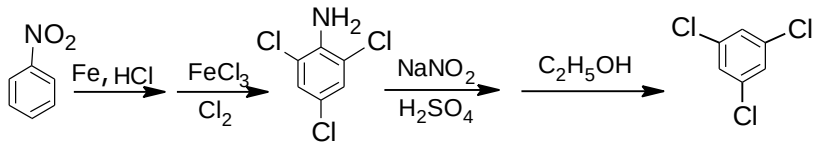
5、 (1) 重氮组分  偶氮组分 

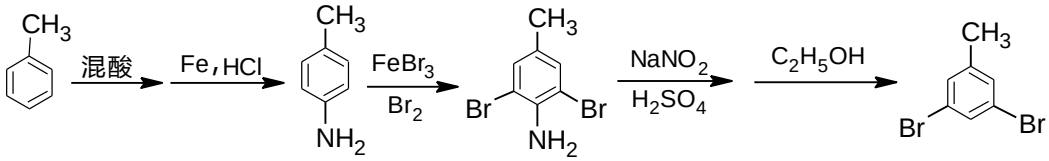
(2) 重氮组分  偶氮组分 

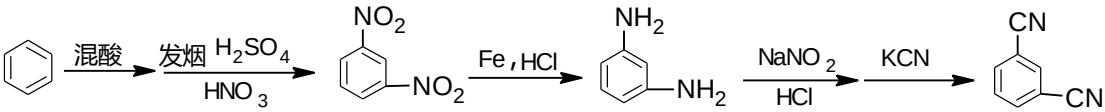
(3) 重氮组分  偶氮组分 

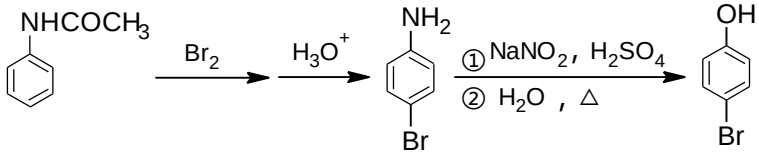
(4) 重氮组分  偶氮组分 

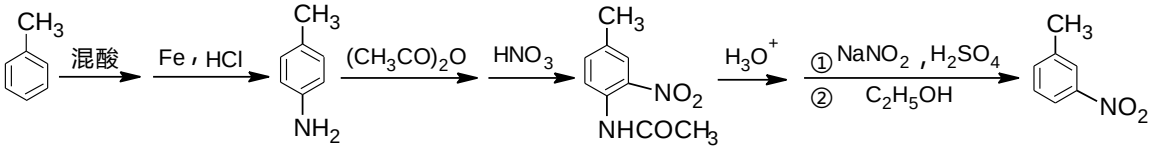
(5) 重氮组分  偶氮组分 

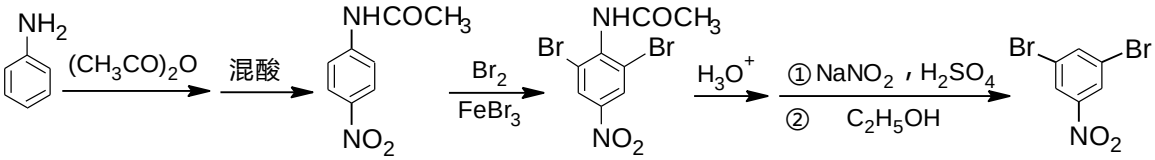
6、 (1) 

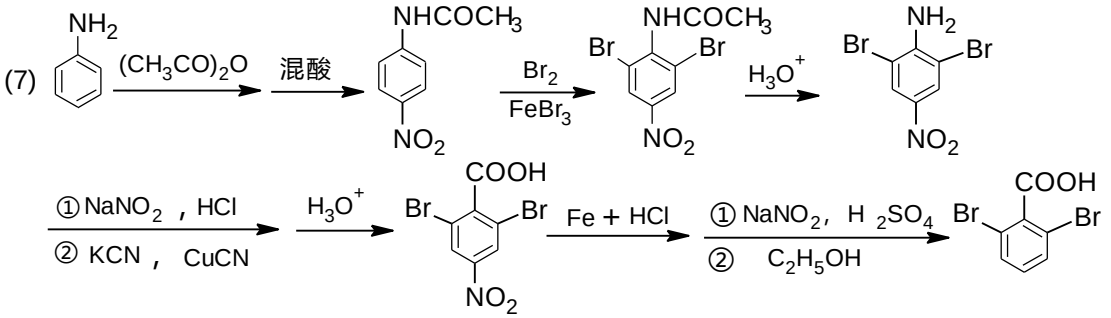
(2) 

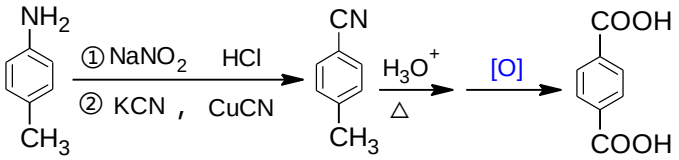
(3) 

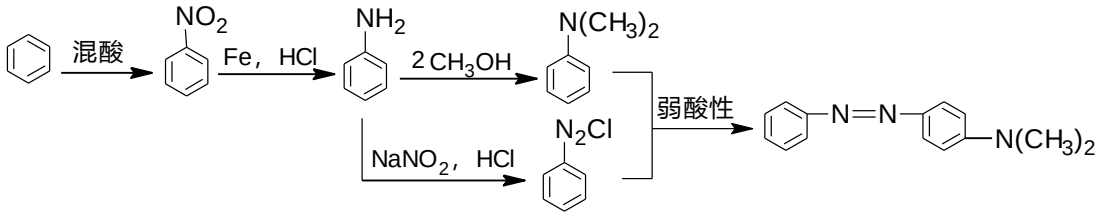
(4) 

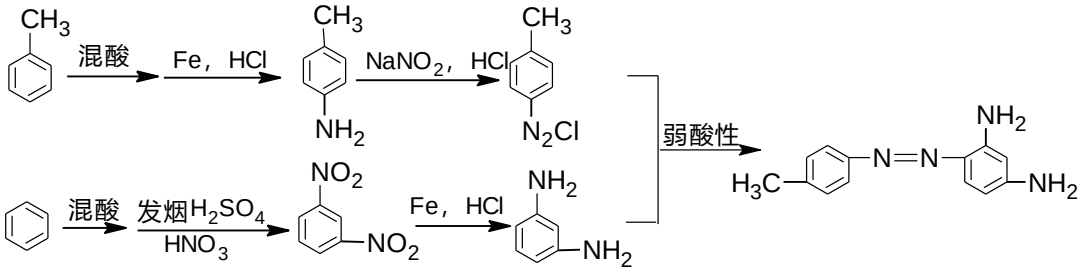
(5) 

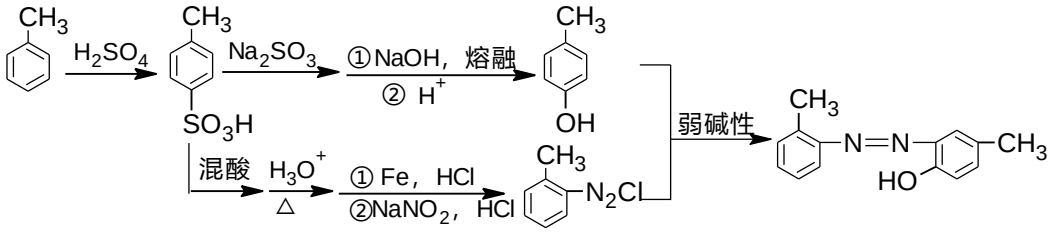
(6) 

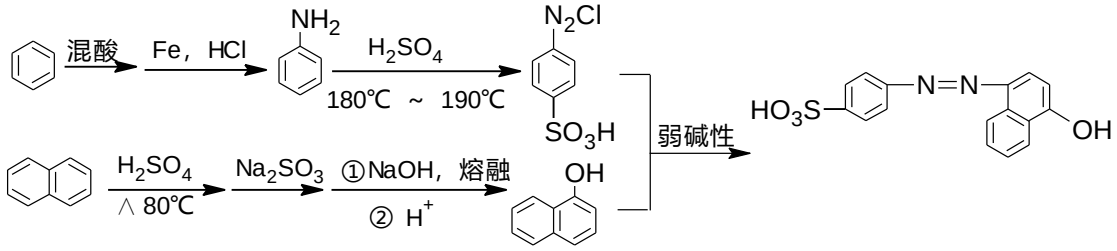
(7) 
NH2
(CH3CO)2O
混酸
NHCOCH3
NO2
Br2
FeBr3
NHCOCH3
Br
Br
NO2
H3O+
NH2
Br
Br
NO2
①NaNO2 ，HCl
② KCN ， CuCN
H3O+
COOH
Br
Br
NO2
Fe + HCl
① NaNO2， H 2SO4
② C2H5OH
COOH
Br
Br

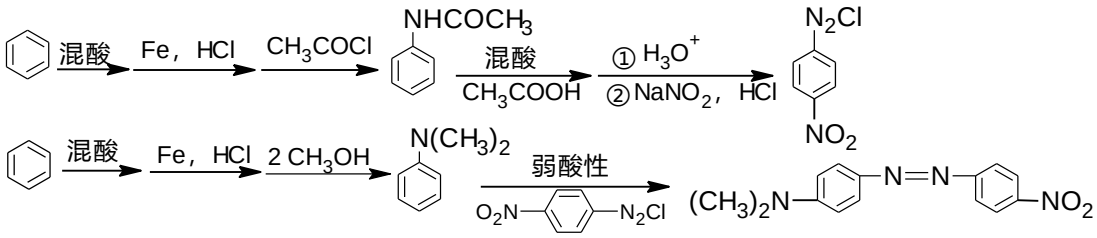
(8) 
NH2
CH3
①NaNO2 HCl
② KCN ， CuCN
CN
CH3
H3O+
△
[O]
COOH
COOH

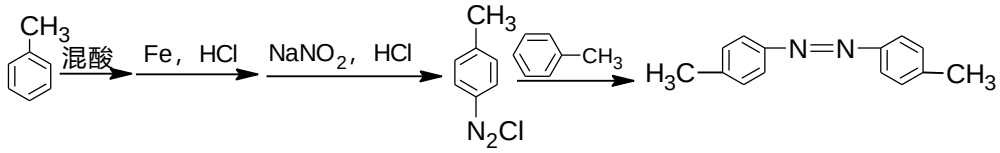
7、(1) 
混酸
NO2
Fe，HCl
NH2
2 CH3OH
N(CH3)2
NaNO2，HCl
N2Cl
弱酸性
N=N
N(CH3)2

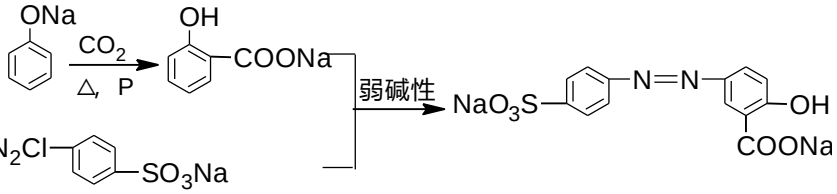
(2) 
CH3
混酸
Fe，HCl
CH3
NH2
NaNO2，HCl
CH3
N2Cl
混酸
发烟H2SO4
HNO3
NO2
NO2
Fe，HCl
NH2
NH2
弱酸性
H3C
N=N
NH2
NH2

(3) 
CH3
H2SO4
CH3
SO3H
Na2SO3
①NaOH，熔融
② H+
CH3
OH
混酸
H3O+
△
① Fe，HCl
②NaNO2，HCl
CH3
N2Cl
弱碱性
CH3
N=N
CH3
HO

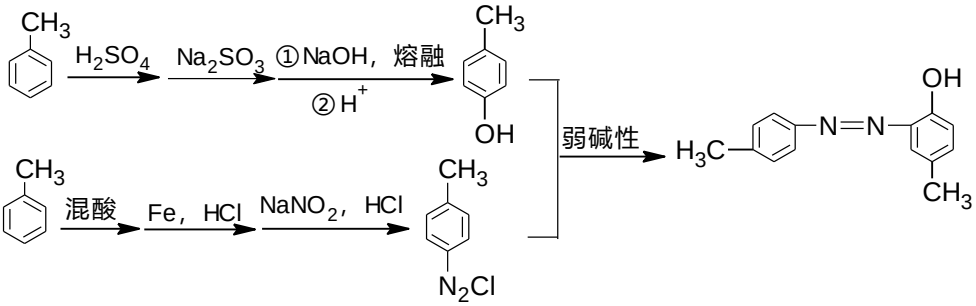
(4) 
混酸
Fe，HCl
NH2
H2SO4
180℃ ~ 190℃
N2Cl
SO3H
H2SO4
∧ 80℃
Na2SO3
①NaOH，熔融
② H+
OH
弱碱性
HO3S
N=N
OH

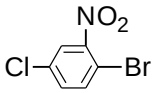
(5) 

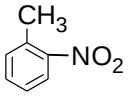
(6) 

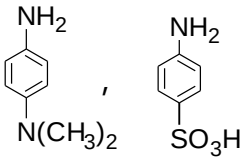
(7) 由苯磺化碱熔融得 

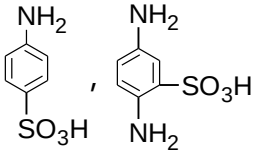
由苯通过(4)法得

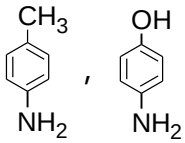
(8) 

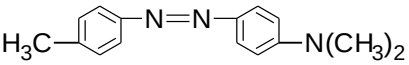
8、 (1) 

(2) 

9、 (1) 

(2) 

(3) 

10ÃB F 8(™ _ Ö 

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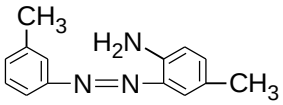
11ÃB F 8(™ _ Ö 

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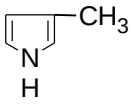
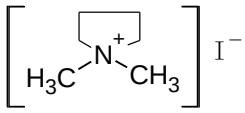
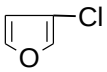
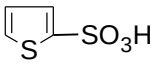
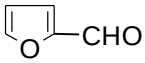
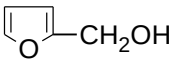
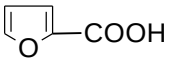
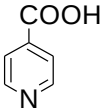
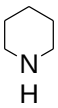
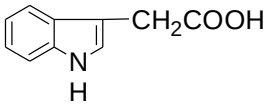
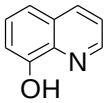
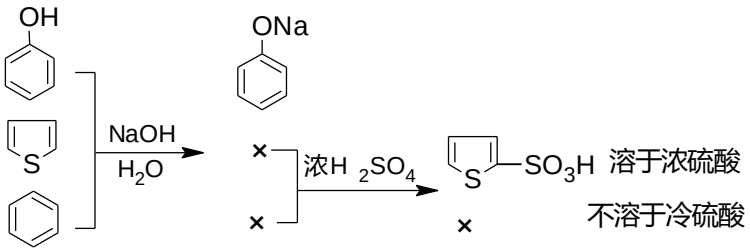
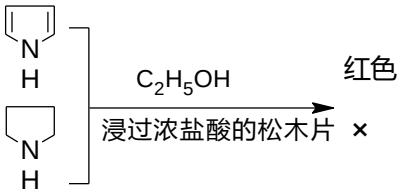
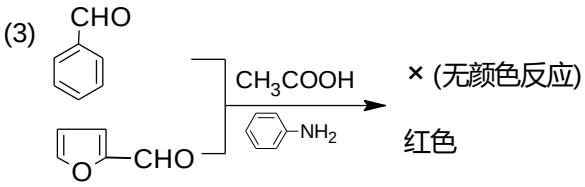
1、 (1)  (2)  (3) (4) 

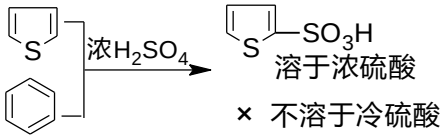
(5)  (6)  ,  , 

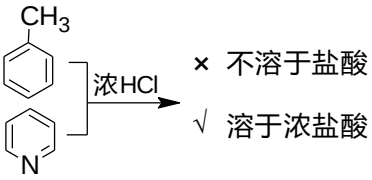
(7)  (8)  (9)  (10) 

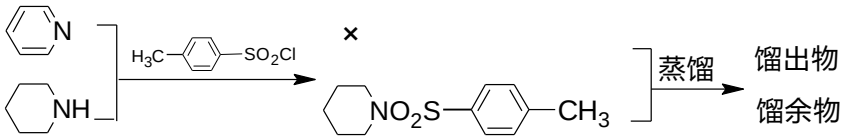
2、 (1) 

(2) 

(3) 

3、 (1) 

(2) 

(3) 

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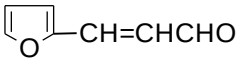
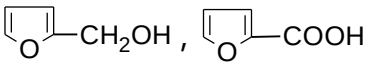
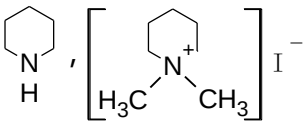
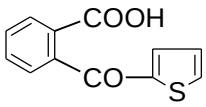
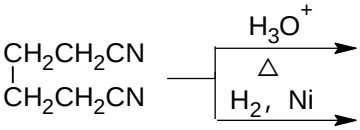
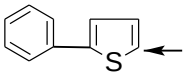
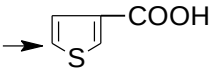
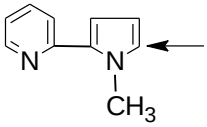
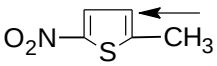
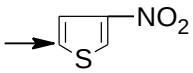
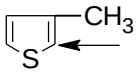
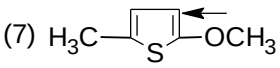
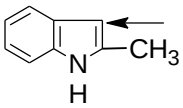
5、 (1)  (2)  (3) 

(4)  (5) H_2，Pt ；HCl ； 

6、 (1)  (2)  (3)  (4) 

(5)  (6)  (7)  (8) 

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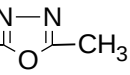
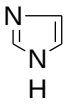
,  ,  ,

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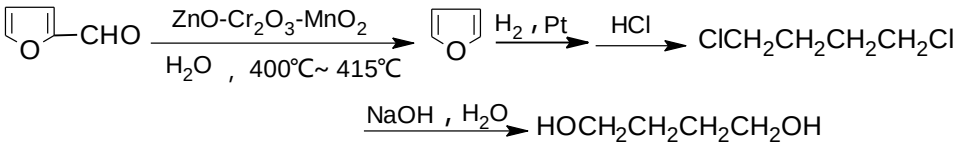
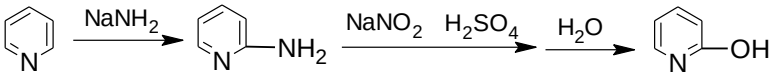
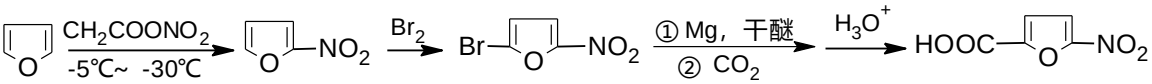
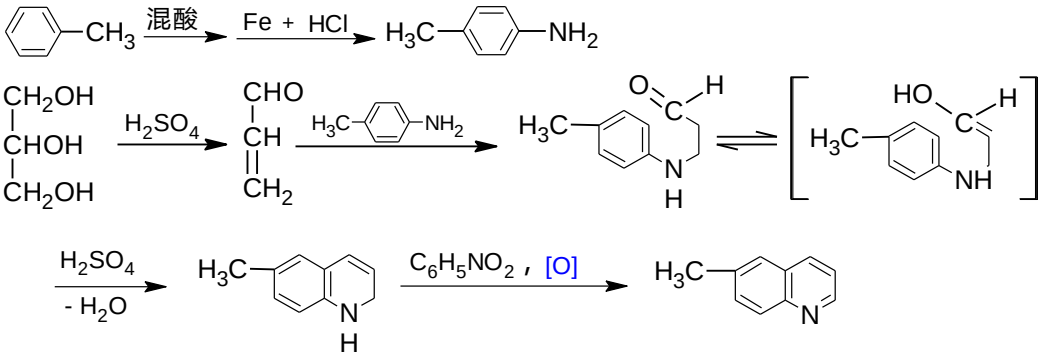
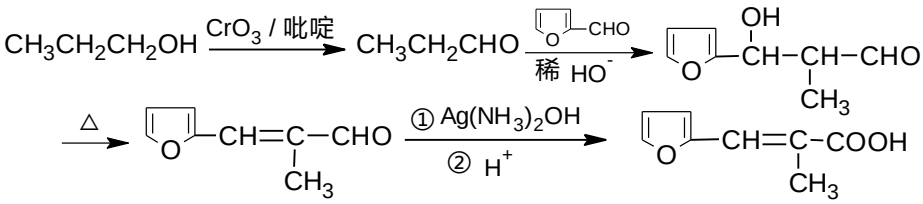
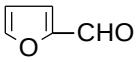
10、(1) 

(2) 

(3) 

(4) 

(5) 

11 Ã Ï •,´ $C_5H_4O_2$,´5 ´ _ ; 

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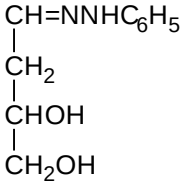
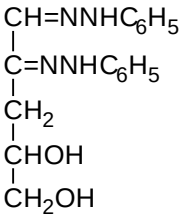
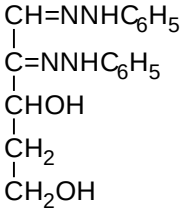
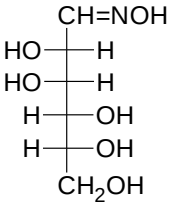
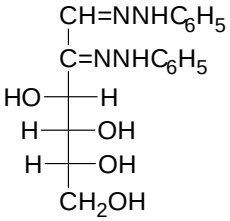
6、 (A) 

(B) 

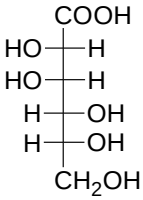
(C) 

7、 (1) 

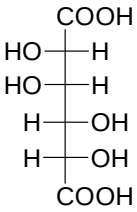
D-甘露糖肟

(2) 

D-甘露糖脎

(3) 

D-甘露糖酸

(4) 

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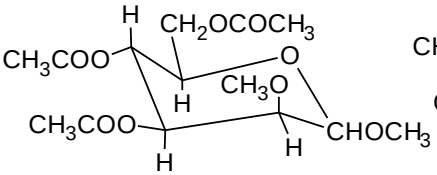
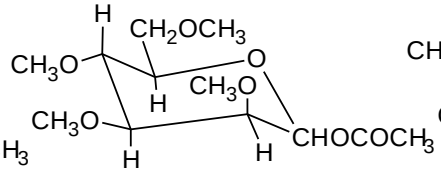
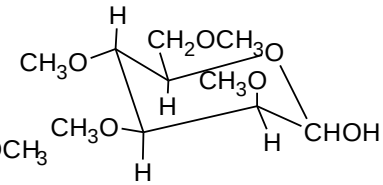


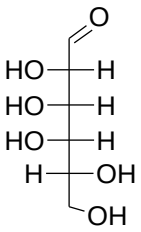
(5) 五乙酸-D-甘露糖酯



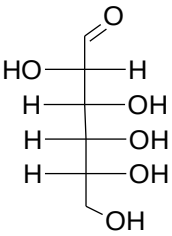
(6) 五甲基-D-甘露糖



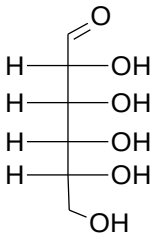
(7) 四甲基-D-甘露糖

8、 

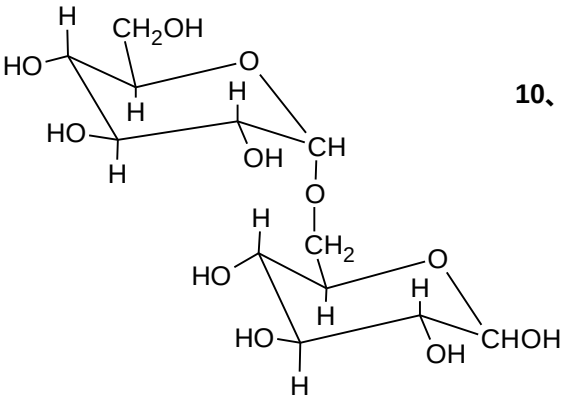
(A) D-太罗糖

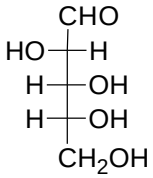


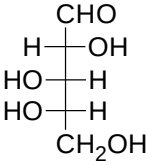
(B) D-阿卓糖

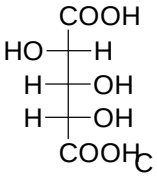


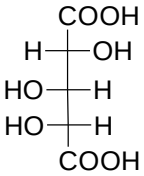
(C) D-阿洛糖

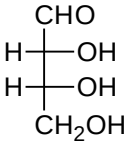
9、 

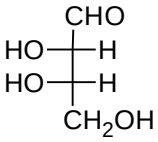
10、 A 为 

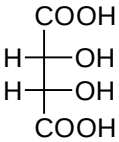
或 

B 为 

或 

C 为 

或 

D 为 

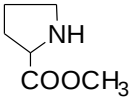
1\ q •1 "X *G¨ Ã<û,-CX ¼ hG¨

1、 (1) $H_2NCH_2CH_2CH_2CH_2\underset{\displaystyle NH_2}{\underset{|}{C}}HCONHCH_2COOH$

(2) $HOOCCH_2CH_2\underset{\displaystyle NH_2}{\underset{|}{C}}HCONH\underset{\displaystyle CH_2CH_2COOH}{\underset{|}{C}}HCONH\underset{\displaystyle CH_2-C_6H_4-OH}{\underset{|}{C}}HCOOH$

(3) $CH_3\underset{\displaystyle NH_2}{\underset{|}{C}}HCONH\underset{\displaystyle CH(CH_3)_2}{\underset{|}{C}}HCONH\underset{\displaystyle CH_2-C_6H_5}{\underset{|}{C}}HCONH\underset{\displaystyle CH_2CH(CH_3)_2}{\underset{|}{C}}HCOOH$

2、 $CH_3\underset{\displaystyle NH_2}{\underset{|}{C}}HCOOH$ $CH_3\underset{\displaystyle HO}{\underset{|}{C}}H\underset{\displaystyle NH_2}{\underset{|}{C}}HCOOH$ $C_6H_5CH_2\underset{\displaystyle NH_2}{\underset{|}{C}}HCOOH$ $\begin{array}{l} S-CH_2-\underset{\displaystyle NH_2}{\underset{|}{C}}HCOOH \\ \underset{\displaystyle S-CH_2-\underset{\displaystyle NH_2}{\underset{|}{C}}HCOOH}{|} \end{array}$ H_2NCH_2COOH

3、 (1) $H_3C-\underset{\displaystyle NHCOCH_3}{\underset{|}{C}}HCOOH$ 溶液有明显的酸性，它可溶于碱而不溶于酸 (2)  溶液有明显的碱性，它可溶于酸而不溶于碱

(3) Äb Å 9 > nG¨ W ×Ä d Å ø > n.á W × ‰ ; Ä a Å ¼ Äc Å ¦] Ä c Å > HNO_3 Œ+X 9N_2 Ä

4、 (1) $CH_3\underset{\displaystyle NH_2}{\underset{|}{C}}HCOO^-$ (2) $C_6H_5CH_2\underset{\displaystyle NH_3^+}{\underset{|}{C}}HCOOH$

5、 (1) $CH_3CH_2CH_2\underset{\displaystyle NH_2}{\underset{|}{C}}HCOOH$，$H_2NCH_2COOH$，$HOOCCH_2\underset{\displaystyle NH_2}{\underset{|}{C}}HCOOH$，$NH_3$

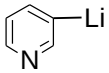
(2) CH_3COOH，H_2NCH_2COOH (3) $H_2NCH_2CH_2CH_2CH_2COOH$

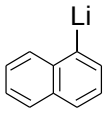
1\ ¼ q1 s3P 9 j F 8(™

1 Ã +• Ä

2 Ã Ä1 Å)+b *9 *K2 Ä2 Å ¼ ‰ *7Ö Ä3 Å.È F ‰ *J+

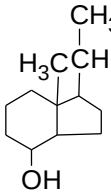
Ä4 Å ¼9 * ¼"_+b.u" Ä5 Å 9!“ I *K

3、 (1)  (2) $(CH_3)_4Si$ (3) $(C_2H_5)_3SiH$ (4) CH_3SiH_2Cl (5) $(C_2H_5)_2Si(OC_2H_5)_2$

4、 (1)  + $CH_3CH_2CH_2Br$ (2) $CH_3CH_2CH_3$ + LiOH (3) $CH_3CH_2CH_3$ + C_2H_5OLi

(4) $CH_3CH(OLi)CH_2CH_2CH_3$; $CH_3CH(OH)CH_2CH_2CH_3$ (5) $CH_3CH_2CH{=}C(O^-){-}CH_3$; $CH_3CH_2CH_2COCH_3$

(6) $(C_2H_5)_2Hg + 2(C_2H_5)_2AlCl$ (7) $(C_2H_5)_3Sb + AlX_3$ (8) $CH_2{=}P(C_6H_5)_3$

(9) $(C_6H_5)_3P{=}CHCH_2CH(CH_3)_2$; 

; H_2, Pt

5、 (1) $CH_3CH_2CH_2CH_2Li + CO_2 \longrightarrow CH_3CH_2CH_2CH_2COOLi \xrightarrow{CH_3CH_2CH_2CH_2Li}$

$CH_3CH_2CH_2CH_2C(OLi)_2CH_2CH_2CH_2CH_3 \xrightarrow{H_2O} CH_3CH_2CH_2CH_2COCH_2CH_2CH_2CH_3$

(2) $CH_3CH_2OH \xrightarrow{HBr} CH_3CH_2Br \xrightarrow[\text{干醚}]{Mg} CH_3CH_2MgBr \xrightarrow{H_3O^+} CH_3CH_2CH(OH)CH_3$

$CH_3CH_2OH \xrightarrow[\text{吡啶}]{CrO_3} CH_3CHO \xrightarrow{[O]} CH_3CH_2COCH_3$

$C_6H_6 + Br_2 \xrightarrow{FeBr_3} C_6H_5Br \xrightarrow[\text{干醚}]{Mg} C_6H_5MgBr \xrightarrow{PCl_3} (C_6H_5)_3P$

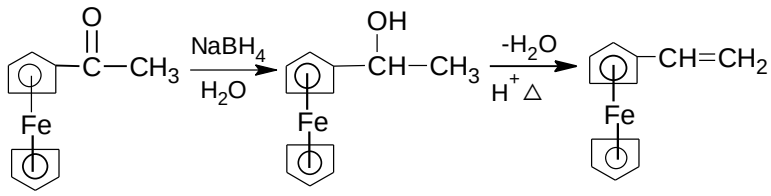
$CH_3CH_2OH \xrightarrow{[O]} CH_3COOH \xrightarrow[H^+]{C_2H_5OH} CH_3COOC_2H_5 \xrightarrow[P]{Br_2} BrCH_2COOC_2H_5$

$\xrightarrow{(C_6H_5)_3P} \xrightarrow{C_2H_5ONa} (C_6H_5)_3P^+{-}\overset{-}{C}HCOOC_2H_5 \xrightarrow{CH_2CH_2COCH_3} CH_3CH_2C(CH_3){=}CHCOOC_2H_5$

(3) 同(2)的方法合成 $(C_6H_5)_3P$

$C_6H_6 + CH_3CH_2Cl \xrightarrow{AlCl_3} C_6H_5CH_2CH_3 \xrightarrow{NBS} C_6H_5CH(Br)CH_3 \xrightarrow{(C_6H_5)_3P}$

$\xrightarrow{C_2H_5ONa} (H_5C_6)_3P^+{-}C(CH_3)^-{-}C_6H_5 \xrightarrow{CH_3COCH_3} (H_3C)_2C{=}C(CH_3)C_6H_5$

(4) $C_2H_5OH \xrightarrow{[O]} CH_3COOH \xrightarrow{SOCl_2} CH_3COCl$

$(C_5H_5)_2Fe + CH_3COCl \longrightarrow$